

data_processing_bb

August 6, 2024

1 Early Data Processing to Understand the dataset.

2 Notebook Information

This notebook is an initial stab at looking at the aquired datasets, seeing what data we have, and seeing how the data can be merged and aligned for future modeling work.

```
[ ]: # Standard package imports
```

```
# Data anaysis
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Plotting and Correlation Maths
```

```
import seaborn as sns
```

```
import scipy as sci
```

```
# Simple model development
```

```
import sklearn as sk
```

```
import os
```

```
[ ]: # You will probably be in the /notebooks directory if you've placed you
```

```
# Jupyter notebook in the right spot.
```

```
os.getcwd()
```

```
[ ]: 'c:\\Users\\btb51\\Documents\\GitHub\\DAAN881_Team1\\opioid_analytics\\notebooks'
```

3 Additional Dataset we probably don't need but I'll drop the link here:

<https://www.datafiles.samhsa.gov/data-sources>

4 Reading data and looking at some summary statistics

4.1 data-table

This seems to include the State, the Data Reported, and the Years Available.

This may not actually be useful.

```
[ ]: dt = pd.read_csv('../data/raw/data-table.csv')
      dt.head(5)
```

```
[ ]:
      State                                Data reported \
0    Alabama                                Unfunded State
1    Alaska  Emergency Department & Inpatient Hospitalization
2    Arizona                                Unfunded State
3    Arkansas                                Unfunded State
4    California  Emergency Department Only

      Years available
0                NaN
1          2018-2022
2                NaN
3                NaN
4          2018-2022
```

```
[ ]: dt.describe()
```

```
[ ]:
      State  Data reported  Years available
count      50             50             25
unique      50             4              2
top    Alabama  Unfunded State  2018-2022
freq         1             25             24
```

4.2 County Opioid Dispensing Rates

Data provided by the CDC at: <https://www.cdc.gov/overdose-prevention/data-research/facts-stats/opioid-dispensing-rate-maps.html>

```
[ ]: count_disp_rates_df = pd.read_csv('../data/raw/County Opioid Dispensing Rates.
      ↪csv')
      count_disp_rates_df.head(5)
```

```
[ ]:
      FullGeoName  YEAR  STATE_NAME  STATE_ABBREV  COUNTY_NAME \
0  AL, Autauga    2019    Alabama             AL  Autauga County
1  AL, Baldwin  2019    Alabama             AL  Baldwin County
2  AL, Barbour  2019    Alabama             AL  Barbour County
3    AL, Bibb   2019    Alabama             AL    Bibb County
4  AL, Blount   2019    Alabama             AL    Blount County
```

	STATE_COUNTY_FIP_U	opioid_dispensing_rate \
0	1001	102.8
1	1003	67.9
2	1005	27.6
3	1007	21.0
4	1009	23.8

	Opioid Dispensing Rate (per 100 persons)
0	>51.0
1	>51.0
2	18.6 - 32.2
3	18.6 - 32.2
4	18.6 - 32.2

```
[ ]: sum(count_disp_rates_df['COUNTY_NAME'] == 'Allegheny')
```

```
[ ]: 0
```

```
[ ]: # What years do we have county data for?
count_disp_rates_df['YEAR'].unique()
```

```
[ ]: array([2019, 2020, 2021, 2022], dtype=int64)
```

```
[ ]: count_disp_rates_df.describe()
```

```
[ ]:
```

	YEAR	STATE_COUNTY_FIP_U	opioid_dispensing_rate
count	12340.000000	12340.000000	12331.000000
mean	2020.498298	30329.250729	38.306496
std	1.118513	15176.674079	30.690663
min	2019.000000	1001.000000	0.000000
25%	2019.000000	18171.000000	18.600000
50%	2020.000000	29147.000000	32.200000
75%	2021.000000	45069.500000	51.000000
max	2022.000000	56045.000000	569.100000

4.3 State Opioid Dispensing Rates

Data provided by the CDC at: <https://www.cdc.gov/overdose-prevention/data-research/facts-stats/opioid-dispensing-rate-maps.html>

```
[ ]: state_disp_rates_df = pd.read_csv('../data/raw/State Opioid Dispensing Rates.
↳ csv')
state_disp_rates_df.head(5)
```

```
[ ]:
```

	YEAR	STATE_NAME	STATE_ABBREV	STATE_FIPS	opioid_dispensing_rate \
0	2019	Alabama	AL	1	86.0
1	2019	Alaska	AK	2	39.3
2	2019	Arizona	AZ	4	44.2

3	2019	Arkansas	AR	5	81.1
4	2019	California	CA	6	31.0

Opioid Dispensing Rate (per 100 persons)

0	>52.8
1	36.1 - 42.7
2	42.8 - 52.8
3	>52.8
4	<36.1

```
[ ]: # What Years do we have state data for?
state_disp_rates_df['YEAR'].unique()
```

```
[ ]: array([2019, 2020, 2021, 2022], dtype=int64)
```

```
[ ]: state_disp_rates_df.describe()
```

```
[ ]:
count      YEAR  STATE_FIPS  opioid_dispensing_rate
count    204.000000   204.000000             204.000000
mean     2020.500000    28.960784             45.566176
std         1.120784    15.715401             12.663324
min      2019.000000     1.000000             24.300000
25%      2019.750000    16.000000             36.175000
50%      2020.500000    29.000000             42.750000
75%      2021.250000    42.000000             52.825000
max      2022.000000    56.000000             86.000000
```

4.4 Dose DX Dashboard information

Data provided by the CDC at: <https://www.cdc.gov/overdose-prevention/data-research/facts-stats/dose-dashboard-nonfatal-discharge-data.html>

Additional information about DOSE can be found at: <https://www.cdc.gov/overdose-prevention/data-research/facts-stats/about-dose-system.html>

TODO I need to go through and get this sheet by sheet. The first sheet I've been reading is 'Data Dictionary' which just talks about what the data is.

```
[ ]: dose_df = pd.read_excel('../data/raw/dose_dx_dashboard_v6.xlsx')
dose_df
# dose_data = pd.read_excel('../data/raw/dose_dx_dashboard_v6.xlsx',
# engine='calamine')
# dose_data
# cols = dose_data[0]
# vals = dose_data[1:]

# dose_df = pd.DataFrame(vals, columns=cols)

# dose_df
```

```

[ ]:
0      US State Submission Counts
1      dataset
2      month
3      year
4      jurisdiction_count
5      State Counts & Rates
6      state
7      month
8      year
9      time_frame
10     dataset
11     count_alldrug
12     count_opioid
13     count_heroin
14     count_stimulant
15     population
16     rate_alldrug
17     rate_opioid
18     rate_heroin
19     rate_stimulant
20     County Counts & Rates
21     state
22     county
23     fips
24     year
25     dataset
26     count_alldrug
27     population
28     rate_alldrug
29     type
30     person_years
31     Overall by Sex & Age
32     state
33     sex
34     age
35     month
36     year
37     time_frame
38     dataset
39     population
40     count_alldrug
41     count_opioid
42     count_heroin
43     count_stimulant
44     rate_alldrug
45     rate_opioid

```

46 rate_heroin
 47 rate_stimulant
 48 * Counts are suppressed when based on 1-9 over...
 49 § Mid-year annual population denominators were...
 50 † Valdez-Cordova Census Area, Alaska was aboli...

Definition \
 0 This sheet provides information on the number ...
 1 Indicator for the data source for that row of ...
 2 All (annual) or month value ranging from 1-12 ...
 3 Year of analysis for that row of data.
 4 Count of states reporting ED visits or Inpatie...
 5 This sheet provides information on the monthly...
 6 The two-digit state abbreviation.
 7 All (annual) or month value ranging from 1-12 ...
 8 Year of analysis for that row of data.
 9 Time frame (annual or monthly) for that row of...
 10 Indicator for the data source for that row of ...
 11 Annual or monthly count of ED visits or inpati...
 12 Annual or monthly count of ED visits or inpati...
 13 Annual or monthly count of ED visits or inpati...
 14 Annual or monthly count of ED visits or inpati...
 15 State population estimates for that row of data.§
 16 Annual or monthly rate of ED visits or inpatie...
 17 Annual or monthly rate of ED visits or inpatie...
 18 Annual or monthly rate of ED visits or inpatie...
 19 Annual or monthly rate of ED visits or inpatie...
 20 This sheet provides information on county-leve...
 21 The two-digit state abbreviation.
 22 County name.
 23 Five-character FIPS code for the county within...
 24 Year of analysis for that row of data.
 25 Indicator for the data source for that row of ...
 26 The number of ED visits for nonfatal all drug ...
 27 County population for that row of data.§ Value...
 28 1-year (annual) or 5-year (2018 2022) rate of ...
 29 Rate type for that row of data; 1 year (annual...
 30 The sum of the mid-year county-level populatio...
 31 This sheet provides information on the monthly...
 32 The two-digit state abbreviation; data on this...
 33 Two sex categories are presented: M = male and...
 34 5 age categories are presented: 0-14, 15-24, 2...
 35 All (annual) or month value ranging from 1-12 ...
 36 Year of analysis for that row of data.
 37 Time frame (annual or monthly) for that row of...
 38 Indicator for the data source for that row of ...
 39 State population estimates for all states subm...

40 Annual or monthly count of ED visits or inpati...
 41 Annual or monthly count of ED visits or inpati...
 42 Annual or monthly count of ED visits or inpati...
 43 Annual or monthly count of ED visits or inpati...
 44 Annual or monthly rate of ED visits or inpatie...
 45 Annual or monthly rate of ED visits or inpatie...
 46 Annual or monthly rate of ED visits or inpatie...
 47 Annual or monthly rate of ED visits or inpatie...
 48 NaN
 49 NaN
 50 NaN

Variable Values

0 NaN
 1 Character (ED or HOSP)
 2 Character (1-12 or all) (where 1=January and 1...
 3 Numeric (2018, 2019, 2020, 2021 or 2022)
 4 Numeric
 5 NaN
 6 Character (e.g., AK=Alaska, US=overall for par...
 7 Character (1-12 or all) (where 1=January and 1...
 8 Numeric (2018, 2019, 2020, 2021 or 2022)
 9 Character (annual or monthly)
 10 Character (ED or HOSP)
 11 Character (count value or suppressed)*
 12 Character (count value or suppressed)*
 13 Character (count value or suppressed)*
 14 Character (count value or suppressed)*
 15 Numeric
 16 Character (rate value or suppressed)*
 17 Character (rate value or suppressed)*
 18 Character (rate value or suppressed)*
 19 Character (rate value or suppressed)*
 20 NaN
 21 Character (e.g., AK=Alaska)
 22 Character†
 23 Character (e.g., 12345, 01234)
 24 Character (2018, 2019, 2020, 2021, 2022 or 201...
 25 Character (ED)
 26 Character (count value or suppressed)*
 27 Numeric
 28 Character (rate value or suppressed)*
 29 Character (1 year or 5 years)
 30 Numeric
 31 NaN
 32 Character (US=overall for participating jurisd...
 33 Character (M = male, F = female, Missing)

```

34 Character (0-14, 15-24, 25-34, 35-54, and 55+ ...
35 Character (1-12 or all) (where 1=January and 1...
36         Numeric (2018, 2019, 2020, 2021 or 2022)
37         Character (annual or monthly)
38         Character (ED or HOSP)
39         Character (count or not available)
40         Character (count value or suppressed)*
41         Character (count value or suppressed)*
42         Character (count value or suppressed)*
43         Character (count value or suppressed)*
44 Character (rate value, suppressed, or not avai...
45 Character (rate value, suppressed, or not avai...
46 Character (rate value, suppressed, or not avai...
47 Character (rate value, suppressed, or not avai...
48                                     NaN
49                                     NaN
50                                     NaN

```

```
[ ]: dose_df.describe()
```

```

[ ]:      Variable      Definition \
count      51      48
unique      28      35
top      year  Year of analysis for that row of data.
freq      4      4

      Variable Values
count      44
unique      19
top      Character (count value or suppressed)*
freq      9

```

4.5 Weather Events

TODO Where did we get this data?

Question This dataset seems to be highly detailed to the Western PA region. We may need to move our analysis from the state of PA, to the Allegheny County (Pittsburgh) region.

```

[ ]: # This is about a 4 minute read on my system (BB) if using just pandas
# Using the python-calamine engine in pandas 2.2.*
# via the python-calamine package (in environment.yaml) speeds this up
# to just 30 seconds. I would recommend installing python-calamine.

weather_events_df = pd.read_excel('../data/raw/WeatherEvents_Jan2016-Dec2022.
↪xlsx', engine='calamine')
weather_events_df.head(5)

```

```
[ ]: EventId Type Severity      StartTime(UTC)      EndTime(UTC) \
0      W-1  Snow      Light  2016-01-06 23:14:00  2016-01-07 00:34:00
1      W-2  Snow      Light  2016-01-07 04:14:00  2016-01-07 04:54:00
2      W-3  Snow      Light  2016-01-07 05:54:00  2016-01-07 15:34:00
3      W-4  Snow      Light  2016-01-08 05:34:00  2016-01-08 05:54:00
4      W-5  Snow      Light  2016-01-08 13:54:00  2016-01-08 15:54:00

      Precipitation(in)      TimeZone AirportCode LocationLat LocationLng \
0              0.00  US/Mountain      K04V      38.0972      -106.1689
1              0.00  US/Mountain      K04V      38.0972      -106.1689
2              0.03  US/Mountain      K04V      38.0972      -106.1689
3              0.00  US/Mountain      K04V      38.0972      -106.1689
4              0.00  US/Mountain      K04V      38.0972      -106.1689

      City      County State ZipCode
0  Saguache  Saguache    CO  81149.0
1  Saguache  Saguache    CO  81149.0
2  Saguache  Saguache    CO  81149.0
3  Saguache  Saguache    CO  81149.0
4  Saguache  Saguache    CO  81149.0
```

```
[ ]: # This only has 45 of the 50 states
print(f'There are {len(weather_events_df['State'].unique())} out of 50 states.')
weather_events_df['State'].unique()
```

There are 45 out of 50 states.

```
[ ]: array(['CO', 'OK', 'MN', 'LA', 'WI', 'ID', 'MI', 'KS', 'WY', 'MA', 'MO',
          'NM', 'NC', 'SC', 'RI', 'VA', 'CT', 'OR', 'ND', 'CA', 'NY', 'OH',
          'SD', 'AZ', 'NV', 'IA', 'TX', 'GA', 'NE', 'TN', 'AL', 'IL', 'AR',
          'WA', 'IN', 'UT', 'FL', 'WV', 'MS', 'PA', 'ME', 'MD', 'NJ', 'KY',
          'VT'], dtype=object)
```

```
[ ]: weather_events_df.describe()
```

```
[ ]:      Precipitation(in)      LocationLat      LocationLng      ZipCode
count      999999.000000  999999.000000  999999.000000  991683.000000
mean              0.089890      39.453285      -92.545095   52737.512165
std              0.561167       5.132021      13.733891   26410.391978
min              0.000000      24.726300     -123.140800   1201.000000
25%              0.000000      36.010300     -99.321600   31510.000000
50%              0.000000      39.794500     -90.993800   55008.000000
75%              0.050000      43.208000     -82.159100   73703.000000
max             115.600000      48.929000     -68.017900   99362.000000
```

```
[ ]:
```

```
[ ]:
```

4.6 Larger Generic Weather Dataset

```
[ ]: generic_weather_df = pd.read_excel('../data/raw/Allegheny County 2017-01-01 to_
↳2024-05-01.xlsx', engine='calamine')
generic_weather_df.head(5)
```

```
[ ]:
      name      datetime  tempmax  tempmin  temp  feelslikemax \
0  Allegheny County  2017-01-01    43.9    31.4  37.1         40.1
1  Allegheny County  2017-01-02    46.3    37.0  42.6         43.3
2  Allegheny County  2017-01-03    48.6    42.9  45.1         46.3
3  Allegheny County  2017-01-04    44.3    21.1  33.8         37.4
4  Allegheny County  2017-01-05    22.6    19.0  20.6         20.8
```

```
      feelslikemin  feelslike  dew  humidity  ...  solarenergy  uvindex \
0          29.0        33.2  28.3      71.1  ...          8.2         4
1          33.2        38.4  38.9      87.1  ...          1.3         1
2          37.7        41.6  42.6      91.0  ...          0.7         0
3           8.0        23.6  23.4      67.4  ...          6.8         4
4           7.0        11.2  11.5      68.4  ...          6.0         3
```

```
      severerisk      sunrise      sunset  moonphase \
0          NaN  2017-01-01T07:43:20  2017-01-01T17:04:23      0.12
1          NaN  2017-01-02T07:43:25  2017-01-02T17:05:15      0.16
2          NaN  2017-01-03T07:43:27  2017-01-03T17:06:08      0.19
3          NaN  2017-01-04T07:43:28  2017-01-04T17:07:02      0.23
4          NaN  2017-01-05T07:43:26  2017-01-05T17:07:58      0.25
```

```
      conditions \
0  Snow, Rain, Partially cloudy
1      Rain, Partially cloudy
2      Rain, Partially cloudy
3  Snow, Rain, Partially cloudy
4      Snow, Partially cloudy
```

```
      description  icon \
0  Clearing in the afternoon with early morning s...  rain
1  Partly cloudy throughout the day with a chance...  rain
2      Partly cloudy throughout the day with rain.  rain
3  Partly cloudy throughout the day with rain or ...  rain
4      Partly cloudy throughout the day with snow.  snow
```

```
      stations
0  KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
1  KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2  KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
3  KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
4  KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
```

[5 rows x 33 columns]

```
[ ]: generic_weather_df
```

```
[ ]:
      name      datetime  tempmax  tempmin  temp  feelslikemax \
0  Allegheny County  2017-01-01    43.9    31.4  37.1         40.1
1  Allegheny County  2017-01-02    46.3    37.0  42.6         43.3
2  Allegheny County  2017-01-03    48.6    42.9  45.1         46.3
3  Allegheny County  2017-01-04    44.3    21.1  33.8         37.4
4  Allegheny County  2017-01-05    22.6    19.0  20.6         20.8
...
2673 Allegheny County  2024-04-27    79.1    53.0  65.3         79.1
2674 Allegheny County  2024-04-28    82.0    63.2  72.6         81.5
2675 Allegheny County  2024-04-29    82.9    62.6  74.1         82.1
2676 Allegheny County  2024-04-30    72.3    61.1  65.9         72.3
2677 Allegheny County  2024-05-01    79.9    53.7  67.5         79.9
```

```
      feelslikemin  feelslike  dew  humidity  ...  solarenergy  uvindex  \
0          29.0        33.2  28.3      71.1  ...         8.2         4
1          33.2        38.4  38.9      87.1  ...         1.3         1
2          37.7        41.6  42.6      91.0  ...         0.7         0
3           8.0        23.6  23.4      67.4  ...         6.8         4
4           7.0        11.2  11.5      68.4  ...         6.0         3
...
2673        53.0        65.3  50.3      60.5  ...        15.3         7
2674        63.2        72.6  55.3      55.6  ...        20.8        10
2675        62.6        73.9  54.6      52.0  ...        23.1         9
2676        61.1        65.9  56.1      71.0  ...        11.9         6
2677        53.7        67.5  48.6      55.4  ...        24.0         8
```

```
      severerisk      sunrise      sunset  moonphase  \
0          NaN  2017-01-01T07:43:20  2017-01-01T17:04:23    0.12
1          NaN  2017-01-02T07:43:25  2017-01-02T17:05:15    0.16
2          NaN  2017-01-03T07:43:27  2017-01-03T17:06:08    0.19
3          NaN  2017-01-04T07:43:28  2017-01-04T17:07:02    0.23
4          NaN  2017-01-05T07:43:26  2017-01-05T17:07:58    0.25
...
2673        10.0  2024-04-27T06:23:41  2024-04-27T20:12:13    0.63
2674        10.0  2024-04-28T06:22:22  2024-04-28T20:13:15    0.67
2675        10.0  2024-04-29T06:21:04  2024-04-29T20:14:17    0.70
2676        10.0  2024-04-30T06:19:47  2024-04-30T20:15:18    0.74
2677        10.0  2024-05-01T06:18:31  2024-05-01T20:16:20    0.75
```

```
      conditions  \
0  Snow, Rain, Partially cloudy
1  Rain, Partially cloudy
```

```

2          Rain, Partially cloudy
3  Snow, Rain, Partially cloudy
4          Snow, Partially cloudy
...
2673       Rain, Partially cloudy
2674             Partially cloudy
2675             Partially cloudy
2676       Rain, Partially cloudy
2677             Clear

```

	description	icon \
0	Clearing in the afternoon with early morning s...	rain
1	Partly cloudy throughout the day with a chance...	rain
2	Partly cloudy throughout the day with rain.	rain
3	Partly cloudy throughout the day with rain or ...	rain
4	Partly cloudy throughout the day with snow.	snow
...	...	...
2673	Partly cloudy throughout the day with rain cle...	rain
2674	Partly cloudy throughout the day.	partly-cloudy-day
2675	Becoming cloudy in the afternoon.	partly-cloudy-day
2676	Partly cloudy throughout the day with afternoo...	rain
2677	Clear conditions throughout the day.	clear-day

	stations
0	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
1	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
3	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
4	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
...	...
2673	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2674	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2675	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2676	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...
2677	KBTP,KAGC,72512464705,KPIT,72520514762,7252009...

[2678 rows x 33 columns]

```
[ ]: generic_weather_df.columns
```

```
[ ]: Index(['name', 'datetime', 'tempmax', 'tempmin', 'temp', 'feelslikemax',
          'feelslikemin', 'feelslike', 'dew', 'humidity', 'precip', 'precipprob',
          'precipcover', 'preciptype', 'snow', 'snowdepth', 'windgust',
          'windspeed', 'winddir', 'sealevelpressure', 'cloudcover', 'visibility',
          'solarradiation', 'solarenergy', 'uvindex', 'severerisk', 'sunrise',
          'sunset', 'moonphase', 'conditions', 'description', 'icon', 'stations'],
          dtype='object')
```

```
[ ]: gen_weather_sel_cols = [
    'name','datetime','tempmax','tempmin','temp','feelslikemax',
    'feelslikemin','feelslike','dew','humidity','precip','precipprob',
    'precipcover','preciptype','snow','snowdepth','windgust','windspeed',
    ↵
    ↪'winddir','sealevelpressure','cloudcover','visibility','solarradiation','solarenergy',
    'uvindex','severerisk','sunrise','sunset','moonphase'
]
```

```
[ ]: down_sel_generic_weather_df = generic_weather_df[gen_weather_sel_cols]
down_sel_generic_weather_df
```

```
[ ]:
```

	name	datetime	tempmax	tempmin	temp	feelslikemax	\
0	Allegheny County	2017-01-01	43.9	31.4	37.1	40.1	
1	Allegheny County	2017-01-02	46.3	37.0	42.6	43.3	
2	Allegheny County	2017-01-03	48.6	42.9	45.1	46.3	
3	Allegheny County	2017-01-04	44.3	21.1	33.8	37.4	
4	Allegheny County	2017-01-05	22.6	19.0	20.6	20.8	
...	...	...	...	...	...	...	
2673	Allegheny County	2024-04-27	79.1	53.0	65.3	79.1	
2674	Allegheny County	2024-04-28	82.0	63.2	72.6	81.5	
2675	Allegheny County	2024-04-29	82.9	62.6	74.1	82.1	
2676	Allegheny County	2024-04-30	72.3	61.1	65.9	72.3	
2677	Allegheny County	2024-05-01	79.9	53.7	67.5	79.9	

	feelslikemin	feelslike	dew	humidity	...	sealevelpressure	\
0	29.0	33.2	28.3	71.1	...	1021.5	
1	33.2	38.4	38.9	87.1	...	1021.8	
2	37.7	41.6	42.6	91.0	...	1006.3	
3	8.0	23.6	23.4	67.4	...	1007.1	
4	7.0	11.2	11.5	68.4	...	1015.1	
...	...	...	...	...	...	...	
2673	53.0	65.3	50.3	60.5	...	1022.5	
2674	63.2	72.6	55.3	55.6	...	1019.8	
2675	62.6	73.9	54.6	52.0	...	1015.8	
2676	61.1	65.9	56.1	71.0	...	1012.7	
2677	53.7	67.5	48.6	55.4	...	1014.5	

	cloudcover	visibility	solarradiation	solarenergy	uvindex	severerisk	\
0	41.7	9.5	95.6	8.2	4	NaN	
1	87.9	5.5	15.5	1.3	1	NaN	
2	78.7	5.3	7.9	0.7	0	NaN	
3	73.0	9.0	78.9	6.8	4	NaN	
4	83.5	6.4	68.1	6.0	3	NaN	
...	...	...	...	...	...	...	
2673	65.9	9.6	177.0	15.3	7	10.0	
2674	47.9	9.9	241.0	20.8	10	10.0	

2675	27.2	9.9	265.8	23.1	9	10.0
2676	68.1	9.9	140.0	11.9	6	10.0
2677	16.2	9.8	278.9	24.0	8	10.0

	sunrise	sunset	moonphase
0	2017-01-01T07:43:20	2017-01-01T17:04:23	0.12
1	2017-01-02T07:43:25	2017-01-02T17:05:15	0.16
2	2017-01-03T07:43:27	2017-01-03T17:06:08	0.19
3	2017-01-04T07:43:28	2017-01-04T17:07:02	0.23
4	2017-01-05T07:43:26	2017-01-05T17:07:58	0.25
...	...	...	...
2673	2024-04-27T06:23:41	2024-04-27T20:12:13	0.63
2674	2024-04-28T06:22:22	2024-04-28T20:13:15	0.67
2675	2024-04-29T06:21:04	2024-04-29T20:14:17	0.70
2676	2024-04-30T06:19:47	2024-04-30T20:15:18	0.74
2677	2024-05-01T06:18:31	2024-05-01T20:16:20	0.75

[2678 rows x 29 columns]

```
[ ]: gen_weather_sel_cols
```

```
[ ]: down_sel_generic_weather_df[gen_weather_sel_cols[20:30]].describe()
```

```
[ ]:
```

	cloudcover	visibility	sollarradiation	solarenergy	uvindex	\
count	2678.000000	2678.000000	2678.000000	2678.000000	2678.000000	
mean	54.781740	8.912211	125.280358	10.809858	4.795370	
std	29.013607	1.339226	79.305213	6.849332	2.471342	
min	0.000000	1.100000	5.800000	0.500000	0.000000	
25%	29.800000	8.500000	59.050000	5.100000	3.000000	
50%	55.250000	9.500000	110.550000	9.500000	5.000000	
75%	80.475000	9.900000	184.075000	15.900000	7.000000	
max	100.000000	9.900000	345.400000	29.800000	10.000000	

	severerisk	moonphase
count	843.000000	2678.000000
mean	14.827995	0.483342
std	14.298733	0.288079
min	10.000000	0.000000
25%	10.000000	0.250000
50%	10.000000	0.490000
75%	10.000000	0.747500
max	100.000000	0.980000

```
[ ]: down_sel_generic_weather_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2678 entries, 0 to 2677
Data columns (total 29 columns):
```

#	Column	Non-Null Count	Dtype
0	name	2678 non-null	object
1	datetime	2678 non-null	object
2	tempmax	2678 non-null	float64
3	tempmin	2678 non-null	float64
4	temp	2678 non-null	float64
5	feelslikemax	2678 non-null	float64
6	feelslikemin	2678 non-null	float64
7	feelslike	2678 non-null	float64
8	dew	2678 non-null	float64
9	humidity	2678 non-null	float64
10	precip	2678 non-null	float64
11	precipprob	2678 non-null	int64
12	precipcover	2678 non-null	float64
13	preciptype	1648 non-null	object
14	snow	2678 non-null	float64
15	snowdepth	2678 non-null	float64
16	windgust	2678 non-null	float64
17	windspeed	2678 non-null	float64
18	winddir	2678 non-null	float64
19	sealevelpressure	2678 non-null	float64
20	cloudcover	2678 non-null	float64
21	visibility	2678 non-null	float64
22	solarradiation	2678 non-null	float64
23	solarenergy	2678 non-null	float64
24	uvindex	2678 non-null	int64
25	severerisk	843 non-null	float64
26	sunrise	2678 non-null	object
27	sunset	2678 non-null	object
28	moonphase	2678 non-null	float64

dtypes: float64(22), int64(2), object(5)

memory usage: 606.9+ KB

```
[ ]: for col in gen_weather_sel_cols:
      print(f'{col}: {down_sel_generic_weather_df[col].value_counts()}')
```

```
name: name
Allegheny County    2678
Name: count, dtype: int64
datetime: datetime
2017-01-01         1
2021-11-15         1
2021-11-17         1
2021-11-18         1
2021-11-19         1
..
2019-06-13         1
```

```

2019-06-14    1
2019-06-15    1
2019-06-16    1
2024-05-01    1
Name: count, Length: 2678, dtype: int64
tempmax: tempmax
78.4    17
73.9    14
76.6    14
57.8    13
82.4    12
..
26.3    1
19.0    1
19.8    1
59.6    1
72.3    1
Name: count, Length: 666, dtype: int64
tempmin: tempmin
64.9    17
66.8    17
30.8    16
38.9    13
28.1    13
..
74.2    1
72.6    1
13.0    1
70.2    1
17.1    1
Name: count, Length: 633, dtype: int64
temp: temp
70.1    13
65.7    13
71.2    13
73.2    12
71.1    11
..
7.9     1
22.6    1
22.8    1
5.7     1
61.6    1
Name: count, Length: 642, dtype: int64
feelslikemax: feelslikemax
78.4    17
73.9    14
76.6    14

```

```

57.8      13
76.9      12
      ..
100.1     1
98.5      1
82.8      1
89.6      1
72.3      1
Name: count, Length: 756, dtype: int64
feelslikemin: feelslikemin
66.8      17
64.9      17
22.8      12
62.6      12
62.1      12
      ..
46.6      1
60.6      1
15.3      1
57.9      1
38.7      1
Name: count, Length: 718, dtype: int64
feelslike: feelslike
71.3      15
65.7      13
73.6      13
64.6      12
60.2      12
      ..
3.5       1
1.1       1
0.6       1
46.1      1
53.8      1
Name: count, Length: 743, dtype: int64
dew: dew
65.0      13
60.6      12
32.1      10
37.4      10
66.4      10
      ..
21.3      1
7.2       1
-10.8     1
-7.1      1
32.2      1
Name: count, Length: 654, dtype: int64

```

```

humidity: humidity
66.2      16
77.6      16
72.0      15
73.2      14
66.7      13
..
92.1      1
41.8      1
29.0      1
90.9      1
40.7      1
Name: count, Length: 590, dtype: int64
precip: precip
0.000     1224
0.001       66
0.008       34
0.003       33
0.002       32
...
1.934       1
0.254       1
0.881       1
0.106       1
1.526       1
Name: count, Length: 557, dtype: int64
precipprob: precipprob
100      1454
0        1224
Name: count, dtype: int64
precipcover: precipcover
0.00      1224
4.17      198
8.33      151
12.50     137
16.67     133
20.83     117
25.00     115
29.17      93
37.50      71
41.67      70
33.33      67
45.83      52
50.00      42
54.17      36
62.50      28
58.33      24
66.67      21

```

```

70.83      20
75.00      17
83.33      12
79.17      11
91.67      11
87.50      10
100.00     10
95.83       8
Name: count, dtype: int64
preciptype: precipdtype
rain      1223
rain,snow  290
snow      112
freezingrain  10
rain,freezingrain  5
freezingrain,snow  3
rain,snow,ice  2
rain,freezingrain,snow  2
freezingrain,snow,ice  1
Name: count, dtype: int64
snow: snow
0.0      2459
0.1       56
0.2       51
0.3       19
0.4       14
0.6       13
0.5       12
0.7        6
0.8        6
1.0        5
0.9        5
1.1        5
1.5        4
1.2        4
1.4        3
1.6        3
1.3        3
2.9        2
1.9        2
2.7        2
4.1        1
2.2        1
2.4        1
2.1        1
Name: count, dtype: int64
snowdepth: snowdepth
0.0      2267

```

0.1	150
0.2	79
0.3	42
0.4	40
0.5	23
0.6	20
0.7	14
0.8	10
0.9	6
1.5	6
1.4	5
1.3	4
1.1	4
1.6	3
1.0	2
2.4	1
1.2	1
1.9	1

Name: count, dtype: int64

windgust: windgust

20.8	96
21.9	91
18.3	88
19.7	78
23.0	75
..	
48.8	1
30.7	1
45.2	1
45.3	1
47.9	1

Name: count, Length: 341, dtype: int64

windspeed: windspeed

9.9	41
8.1	36
8.3	35
11.9	34
9.0	32
..	
22.1	1
25.1	1
25.6	1
23.0	1
20.4	1

Name: count, Length: 229, dtype: int64

winddir: winddir

244.1	7
220.5	6

```

183.5    6
204.0    5
249.4    5
..
116.5    1
69.0     1
48.9     1
150.1    1
262.8    1
Name: count, Length: 1783, dtype: int64
sealevelpressure: sealevelpressure
1017.9    27
1015.0    24
1017.1    23
1014.5    22
1018.9    22
..
1035.1    1
1000.3    1
1002.4    1
1036.0    1
996.7     1
Name: count, Length: 349, dtype: int64
cloudcover: cloudcover
100.0     74
99.9      15
99.7      15
99.8      12
99.1      12
..
12.6      1
14.2      1
90.0      1
34.2      1
66.2      1
Name: count, Length: 911, dtype: int64
visibility: visibility
9.9       718
9.8       227
9.7       155
9.6       127
9.5       123
...
1.1        1
5.2        1
1.9        1
2.2        1
1.2        1

```

```

Name: count, Length: 77, dtype: int64
solarradiation: solarradiation
105.3    6
44.5     6
75.5     6
77.3     6
14.0     6
..
56.0     1
129.7    1
101.1    1
127.7    1
278.9    1
Name: count, Length: 1689, dtype: int64
solarenergy: solarenergy
2.0      24
3.9      24
6.5      23
8.2      22
6.8      22
..
27.0     1
27.4     1
25.1     1
0.5      1
25.8     1
Name: count, Length: 278, dtype: int64
uvindex: uvindex
4        386
5        340
6        307
1        302
7        290
3        289
8        287
2        274
9        167
10       21
0        15
Name: count, dtype: int64
severerisk: severerisk
10.0     725
30.0      75
60.0      25
75.0      12
100.0      6
Name: count, dtype: int64
sunrise: sunrise

```

```

2017-01-01T07:43:20    1
2021-11-15T07:06:34    1
2021-11-17T07:08:53    1
2021-11-18T07:10:03    1
2021-11-19T07:11:12    1
..
2019-06-13T05:49:16    1
2019-06-14T05:49:14    1
2019-06-15T05:49:15    1
2019-06-16T05:49:17    1
2024-05-01T06:18:31    1
Name: count, Length: 2678, dtype: int64
sunset: sunset
2017-01-01T17:04:23    1
2021-11-15T17:02:24    1
2021-11-17T17:00:52    1
2021-11-18T17:00:08    1
2021-11-19T16:59:27    1
..
2019-06-13T20:51:04    1
2019-06-14T20:51:30    1
2019-06-15T20:51:54    1
2019-06-16T20:52:16    1
2024-05-01T20:16:20    1
Name: count, Length: 2678, dtype: int64
moonphase: moonphase
0.25    105
0.75     99
0.50     91
0.00     90
0.03     35
...
0.77     12
0.74     11
0.24     11
0.76     10
0.26      9
Name: count, Length: 96, dtype: int64

```

```
[ ]: for col in gen_weather_sel_cols:
      print(f'{col}: {down_sel_generic_weather_df[col].unique()}')
```

```

name: ['Allegheny County']
datetime: ['2017-01-01' '2017-01-02' '2017-01-03' ... '2024-04-29' '2024-04-30'
          '2024-05-01']
tempmax: [43.9 46.3 48.6 44.3 22.6 17.6 16.  15.7 27.9 49.  54.7 64.4 36.6 35.6
          34.1 38.7 60.2 51.3 41.7 55.8 63.2 60.8 49.9 39.1 51.9 50.3 31.9 33.
          31.6 37.9 24.9 29.9 44.8 54.4 63.4 55.9 29.6 32.9 54.5 55.7 38.8 48.7]

```

42. 28.4 52.8 65.9 59.5 65.5 65.1 67.9 75.7 65.6 45.4 56.7 61.1 39.8
 30.3 46.2 56. 57. 33.7 25.9 27.7 39.4 35.3 35.2 41. 43.3 39.5 51.
 59.3 41.9 49.5 73.4 72.7 65.4 68.1 59.7 60.1 55.6 52.9 48. 57.1 70.3
 63.1 70.6 64.2 38.2 57.7 71.9 78.4 70. 65.2 71.4 80.8 74.6 67.5 70.1
 75.4 68.5 53.9 66.5 61. 63.6 76.4 80.9 70.4 73.9 85.2 80.6 61.2 58.1
 64.7 71.1 47.8 63.9 70.8 58.6 66.2 68.7 69.9 82.3 86.1 84.5 70.9 74.7
 65.3 71.3 66.8 66.6 77.3 78. 74.9 73.1 74.8 78.9 79.8 73.7 64. 69.8
 76.1 82.4 84.6 86.3 88.7 83.2 84.3 81.2 85.5 87.2 77.6 83.3 75.5 74.3
 73. 74. 84.1 83.1 81. 83. 80.7 77.4 79.5 81.8 87. 76.7 77.8 80.3
 83.9 86.4 77.1 79.4 78.8 79.7 81.4 78.1 82.8 82.5 71.5 74.4 76.5 76.9
 83.8 82.1 84.2 72.1 73.3 77.2 74.5 60.7 65. 80.1 73.6 68.3 64.1 67.4
 68.9 78.2 68.8 72.2 79. 82. 85.1 85.9 88.3 85.4 87.1 89.2 62. 74.2
 70.7 75.8 56.9 69.4 65.8 58.9 45.8 51.6 42.2 47.1 45.6 48.1 62.7 59.
 52. 42.4 37.4 46.7 43.7 44.6 47.6 47. 58.8 42.8 48.5 38.9 52.3 44.7
 55. 61.5 46.9 53.8 42.3 36.2 26.9 34.5 30.8 37.5 39.3 22.8 31.3 26.8
 40.9 39.7 42.9 51.8 40.8 52.2 37.1 32.5 18.8 15.1 13.2 17.9 17.3 12.8
 10.5 12.4 22.4 16.6 8.2 10.8 26.2 33.4 35.8 46.5 23.8 15.3 14.4 25.5
 38.1 62.8 55.1 34.9 28.6 47.2 41.5 44.2 25.2 29.8 38. 28.1 30.6 23.6
 39.9 49.3 34.3 51.1 60.9 33.1 62.6 76.6 57.8 41.2 44.4 39.6 31.1 29.3
 40. 40.5 41.3 34. 29.1 41.1 35.4 31.4 49.1 48.4 50.2 58.5 43.6 37.6
 38.4 33.6 75.1 82.9 57.6 56.8 60.4 50.5 78.6 75.2 64.3 82.2 73.5 81.6
 74.1 77.7 72.5 88.5 59.9 77.9 79.2 89.3 79.6 75.3 72.8 76.8 78.3 79.9
 88.1 90.3 88.4 92. 83.7 85.6 87.3 88.2 78.7 85.7 87.9 80.4 78.5 88.8
 89.7 89.8 91.8 91.7 67.1 67.3 81.7 86. 66.9 67.7 71.6 62.5 63. 67.6
 53. 63.5 51.2 49.6 54.8 56.6 47.5 45.5 49.4 46.1 61.7 51.4 44.1 35.7
 43.5 49.8 39.2 37.3 26.6 45.1 46.6 52.4 50.7 26.1 30.5 42.1 50.4 30.2
 30.7 31. 33.8 46.8 50.9 40.4 45.7 47.9 42.7 34.8 33.3 40.6 42.5 23.9
 28.2 37. 36.4 11.4 25.1 24.2 32.1 41.4 14.7 9.7 18.6 26.3 35.1 46.4
 34.6 30. 46. 53.5 35. 24.7 19. 19.8 29.7 36.3 66.4 55.5 47.4 60.
 56.5 53.6 67.2 51.7 61.6 76.3 70.5 73.2 81.3 84.8 59.6 76. 70.2 71.2
 80. 86.7 89. 88.6 87.4 77. 86.6 89.4 75.6 86.5 79.1 83.6 83.4 81.5
 84.4 79.3 75. 77.5 86.8 82.6 87.8 68. 61.3 66.7 54.9 63.7 71.7 56.3
 55.4 61.8 53.7 36.8 41.6 43.1 52.1 38.3 45. 40.1 27.5 61.9 57.2 57.3
 38.6 41.8 54.3 28.5 26.5 24. 32.4 53.2 35.9 32.7 32. 36.7 34.7 50.1
 65.7 48.3 44. 42.6 69.1 57.5 56.2 56.1 52.5 47.7 58.3 69.3 62.3 50.6
 69.7 84.7 69. 83.5 81.9 80.2 89.5 90.6 92.3 93.7 92.7 91.1 91.2 90.9
 92.1 85. 87.5 88. 84. 85.8 67. 62.4 58.4 72.9 53.3 60.3 55.3 44.9
 71. 50.8 53.4 36. 39. 32.2 43.4 33.9 34.4 25.8 30.4 27.4 28.8 29.4
 28.9 32.8 24.8 28.7 22.7 62.1 36.5 64.6 59.8 64.8 55.2 63.8 72. 53.1
 64.9 69.2 62.9 57.4 69.6 90.4 82.7 85.3 90.2 84.9 73.8 68.4 37.7 38.5
 52.6 45.2 28.3 20. 22.9 19.5 17.1 31.5 18.9 23.7 25.6 21.5 47.3 37.2
 58.7 48.8 60.5 25.4 68.2 66.3 59.1 75.9 91. 87.7 59.2 62.2 59.4 66.
 67.8 31.8 54.1 45.3 40.7 40.3 40.2 12.2 20.6 37.8 33.2 43.2 66.1 89.6
 71.8 90. 80.5 89.1 76.2 81.1 72.4 56.4 52.7 54.6 36.9 19.7 19.2 27.
 24.3 43.8 27.8 58. 50. 44.5 57.9 72.3]
 tempmin: [31.4 37. 42.9 21.1 19. 6.6 7.6 7.1 4.9 23.9 39.7 38.5 28.6 26.8
 24.6 25.7 38.4 38. 34.4 43.5 48.3 33.8 35.3 33.2 28.1 27.7 21.7 22.1
 33.1 24.3 13.8 24.7 28.7 30.5 17.3 14. 34.3 35.8 26.7 26. 22.5 20.8

44.1 44.2 39.2 47.3 52.9 53.6 56.4 29. 26.1 32.9 37.4 43.9 27.2 21.4
 18.5 16.2 37.3 42.4 38.2 36.4 15.1 16.1 18.9 14.4 20.9 21.2 32.4 33.4
 34.5 44.5 25.9 19.3 62. 52.4 52.2 47.5 44.8 48.5 38.9 33.3 54.3 45.
 37.1 32.3 35.6 57.4 57.5 42.1 48.2 51.5 62.8 46.6 56.7 58.2 51.6 42.7
 38.8 46.8 49.1 52. 58. 50. 54.2 58.4 60.4 48.8 45.7 48.7 41.3 41.2
 47.6 49.9 50.1 42.3 45.8 64.3 68.4 64. 59.2 57.8 49.3 57.7 55.4 56.8
 55.7 57. 60.2 61.5 56.3 54.1 54.5 59.9 53.9 51.8 51.2 54. 63.1 66.2
 69.1 67. 68.6 62.9 70.3 58.9 62.3 64.4 67.9 65.6 61.2 55.3 54.7 49.8
 59.7 66.1 73. 67.5 65.1 67.7 56.2 65. 71.3 65.8 61.8 66.7 69.7 70.9
 70.4 72. 68.1 59.3 64.7 64.9 55.9 60.1 63.6 65.7 65.4 66.4 59.5 61.
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6.920e+01	6.290e+01	5.740e+01	6.960e+01	8.630e+01	9.280e+01
9.630e+01	9.070e+01	9.270e+01	8.550e+01	9.890e+01	9.250e+01
9.360e+01	7.380e+01	8.060e+01	6.840e+01	4.640e+01	4.760e+01
3.810e+01	3.780e+01	4.420e+01	2.800e+01	5.260e+01	2.010e+01
2.440e+01	1.030e+01	1.470e+01	1.510e+01	3.960e+01	1.530e+01
1.930e+01	2.060e+01	3.890e+01	1.500e+01	1.880e+01	1.440e+01
2.330e+01	6.600e+00	4.150e+01	3.210e+01	1.760e+01	5.870e+01
2.220e+01	6.050e+01	4.580e+01	2.250e+01	4.940e+01	6.820e+01
6.630e+01	5.910e+01	8.020e+01	7.590e+01	1.006e+02	9.090e+01
8.040e+01	8.990e+01	5.920e+01	6.220e+01	5.940e+01	6.600e+01
6.780e+01	3.450e+01	1.920e+01	5.410e+01	3.190e+01	4.700e+01
2.760e+01	-4.200e+00	1.100e+01	1.800e+01	1.790e+01	2.950e+01
4.650e+01	3.460e+01	4.820e+01	6.610e+01	8.520e+01	7.180e+01
9.320e+01	9.410e+01	9.370e+01	9.210e+01	7.620e+01	7.240e+01
5.640e+01	5.270e+01	4.950e+01	4.590e+01	4.060e+01	4.670e+01
4.070e+01	5.460e+01	3.990e+01	1.770e+01	1.970e+01	2.490e+01
4.450e+01	5.800e+01	5.000e+01	3.510e+01	5.790e+01	7.230e+01]
feelslikemin: [29. 33.2 37.7 8. 7. -2.8 -4.2 -9.1 -5.8 15.1 34.1					
30.8					
21.9	21.	19.5	25.7	34.8	31.3
22.5	18.1	14.5	18.6	12.8	12.7
40.1	21.5	2.2	26.9	24.2	15.6
36.2	43.1	52.9	53.6	56.4	17.2
9.5	32.9	34.7	30.5	30.7	10.2
13.6	26.6	30.4	41.5	16.4	13.
38.9	44.6	54.3	28.2	22.2	23.8
62.8	45.5	56.7	58.2	51.6	39.1
54.2	58.4	60.4	42.4	39.9	44.5
50.1	38.5	64.3	68.4	64.	59.2
57.	60.2	61.5	56.3	54.1	54.5
66.2	69.1	67.	68.6	62.9	70.3
55.3	54.7	49.2	59.7	66.1	73.
65.8	61.8	66.7	69.7	70.9	70.4
60.1	63.6	65.7	65.4	66.4	59.5
63.2	53.3	57.1	60.9	60.6	59.1
57.2	61.6	63.5	62.1	62.2	61.7
46.9	53.1	63.8	63.3	66.8	68.
42.7	50.4	41.3	36.3	32.7	27.8
26.2	28.6	11.8	7.8	25.	30.1
21.8	39.3	32.2	37.5	28.9	23.7
21.7	1.1	-2.3	11.3	29.4	37.1
0.3	-3.6	-9.4	-1.2	-7.6	-10.6

-16.6	19.7	18.4	36.8	12.1	-6.	-3.5	-9.8	9.4	24.4	34.3	41.6	
25.6	16.9	12.9	12.5	31.	18.5	1.7	-1.4	16.	-5.6	-2.9	22.3	
13.9	7.1	33.1	17.	8.3	33.5	52.5	20.4	18.3	23.5	59.4	37.3	
34.4	31.5	23.6	22.7	13.7	14.8	13.5	18.9	11.4	10.4	16.8	10.	
13.8	24.7	20.2	13.3	26.4	21.2	26.7	15.9	22.1	19.1	20.5	30.	
43.5	41.9	23.1	24.	20.7	29.2	40.7	44.7	39.2	45.7	32.4	55.6	
52.1	52.6	52.8	61.1	61.9	64.2	57.3	69.8	63.7	45.9	63.	67.2	
72.1	72.4	66.9	65.2	58.3	68.9	69.3	71.6	70.	56.9	68.8	60.7	
72.9	68.3	59.8	65.9	70.1	68.7	61.3	64.1	67.6	69.2	66.	68.5	
69.9	70.8	71.4	67.1	56.	53.2	51.4	53.7	33.3	38.2	32.6	26.1	
22.9	37.	31.6	33.9	34.	44.4	26.3	29.1	16.1	18.	26.	24.6	
40.3	20.9	9.9	11.2	38.8	9.6	10.9	30.6	22.4	25.9	23.3	25.4	
26.5	23.2	21.1	28.4	25.3	32.1	9.1	9.3	14.9	16.6	15.4	19.4	
-9.5	-13.8	-9.	27.4	0.9	3.5	1.5	-24.1	-22.	-1.3	7.6	17.5	
20.6	15.	8.6	19.	6.9	1.9	-7.1	-6.9	5.8	21.6	24.3	20.8	
35.2	15.2	22.6	35.	50.7	32.3	43.6	45.2	35.1	24.9	46.2	44.3	
49.9	41.2	37.9	52.7	45.	51.7	51.	37.6	63.9	59.	41.8	65.5	
47.6	55.8	66.6	54.6	55.5	69.6	67.8	69.5	69.4	70.6	72.5	75.5	
71.5	66.3	62.7	54.8	56.5	54.4	47.4	60.3	62.5	55.	51.9	43.9	
34.2	42.8	48.4	39.7	38.6	49.3	33.6	6.3	4.3	20.1	27.	21.3	
29.8	28.7	19.8	39.	15.8	7.4	4.6	-0.3	14.7	19.2	45.6	31.1	
26.8	11.7	13.1	2.7	10.7	3.6	21.4	3.2	18.8	4.9	6.5	4.8	
35.9	42.5	32.8	37.8	25.2	36.6	31.8	58.8	36.7	30.3	36.9	39.5	
37.4	40.8	27.5	32.	57.9	46.	66.5	50.3	71.9	58.7	51.3	60.	
58.5	73.1	74.2	72.6	68.2	70.2	64.5	63.4	58.6	45.1	50.5	37.2	
44.9	44.	40.4	58.1	42.	43.4	32.5	35.6	41.1	42.9	46.3	17.8	
19.9	14.	-2.7	-3.2	16.3	11.5	6.2	6.	4.2	2.1	17.1	8.9	
14.6	0.2	12.	17.3	42.2	36.5	13.4	38.1	47.2	40.6	36.1	49.1	
48.3	39.8	48.5	71.	47.5	44.8	73.6	69.	70.5	71.2	71.7	53.5	
51.1	59.6	50.2	27.7	29.3	29.7	30.2	35.7	29.6	9.2	3.3	-0.2	
12.4	3.7	1.8	13.2	0.8	-1.	-9.3	7.5	8.8	-13.7	-4.8	5.	
2.5	16.5	27.6	5.9	2.6	5.6	14.4	17.9	1.2	-2.	27.3	44.1	
53.4	47.1	53.	48.8	57.6	74.4	70.7	39.4	25.1	33.7	6.7	14.1	
-30.4	-25.4	-6.5	-1.7	43.3	11.	-6.1	28.	28.5	10.1	8.1	46.1	
49.5	54.9	75.3	73.9	67.3	46.5	42.3	45.4	35.8	40.5	42.1	28.1	
48.2	-8.3	8.4	-1.9	0.5	5.1	42.6	17.4	43.7	38.7]			
feelslike:	[	33.2	38.4	41.6	23.6	11.2	1.9	1.	-3.5	7.4	27.8	44.5
56.5												
26.3	28.	27.5	32.2	48.9	36.1	34.1	43.5	52.7	54.5	37.6	31.4	
39.9	29.7	19.8	19.4	22.	16.4	24.	28.2	22.2	12.1	13.4	29.3	
37.2	53.3	38.9	9.9	10.5	43.4	42.7	25.	30.9	25.2	17.5	33.3	
52.3	50.9	48.4	54.1	58.5	60.4	65.6	42.6	22.8	33.9	47.2	54.6	
23.3	16.	16.7	24.5	44.1	53.1	43.8	44.4	20.5	9.6	20.9	15.5	
5.3	18.	25.7	32.3	31.	40.6	49.2	22.6	29.8	53.6	66.7	58.7	
59.5	55.	46.6	50.2	35.	43.3	57.2	59.6	54.4	49.	40.9	53.7	
68.1	63.7	55.3	50.7	59.8	66.2	67.3	60.7	64.5	66.3	62.4	46.3	
52.2	53.5	55.2	63.5	68.5	61.	63.	71.7	69.9	57.5	45.7	49.8	
59.2	53.2	52.	55.7	59.	56.1	65.4	75.5	75.4	73.	62.9	60.2	

62.	69.6	63.6	63.8	66.	69.4	67.1	66.5	57.8	65.8	72.3	75.3
78.4	77.4	73.7	74.3	71.8	75.7	76.9	70.9	75.6	71.3	67.8	64.6
61.7	64.1	72.6	76.3	76.2	73.6	74.9	76.5	72.7	69.3	68.4	67.5
72.4	77.	74.8	72.2	78.	81.4	79.	79.7	73.8	74.	71.	67.
70.	65.3	71.4	73.9	74.5	74.2	74.6	73.1	65.7	67.6	68.6	70.1
68.3	69.2	75.8	70.6	70.2	75.1	69.	64.	61.9	66.6	66.8	67.4
56.4	57.3	63.1	68.9	68.	56.	58.3	57.4	57.	62.2	70.8	71.5
73.5	76.7	77.1	76.1	57.7	54.	59.1	63.2	63.9	65.	66.9	50.5
48.8	56.6	60.8	53.	38.6	40.	48.	33.4	33.6	38.5	59.4	37.1
39.7	19.1	20.7	35.8	37.8	35.7	34.9	34.	42.	28.9	26.6	41.9
29.6	35.6	45.9	32.8	37.9	48.1	49.3	42.9	26.5	16.3	21.5	19.2
27.6	6.3	15.2	24.9	31.3	30.	8.1	4.4	0.	6.1	4.	-1.6
-7.7	-6.2	1.3	-1.7	-14.6	-10.	0.6	23.2	26.8	31.2	49.6	41.8
1.1	3.5	14.8	-3.7	2.2	18.8	31.6	51.7	41.4	31.9	16.9	2.1
8.7	9.7	10.2	21.8	38.8	39.1	22.3	24.1	42.3	57.6	28.5	33.8
50.	45.4	50.4	39.3	41.5	49.9	44.2	21.3	22.1	23.7	26.2	17.2
16.2	26.	24.7	19.9	14.9	19.6	34.4	40.3	25.8	20.6	21.4	24.2
26.9	21.6	31.8	52.1	38.7	32.9	44.6	26.4	40.1	26.1	40.5	71.6
58.9	34.8	44.	50.3	47.1	55.5	33.5	62.7	60.5	56.2	59.3	63.4
71.2	65.9	68.7	68.2	64.4	68.8	72.1	73.2	77.6	76.6	75.9	61.3
61.6	84.3	69.8	69.7	71.9	76.4	80.1	82.8	81.	81.7	83.8	67.2
70.3	77.9	80.5	79.2	79.6	69.5	73.3	70.7	76.	79.5	72.5	71.1
72.9	80.8	81.2	82.2	83.1	84.1	77.3	61.4	55.8	74.4	73.4	72.
59.9	60.	67.9	59.7	66.1	45.8	46.8	55.6	40.2	42.4	42.8	44.9
46.	40.8	39.5	38.3	45.2	56.7	42.2	35.4	27.2	23.8	25.3	23.
31.1	36.8	27.4	16.5	35.2	17.	12.6	19.7	51.9	33.	25.4	29.
36.9	34.7	30.4	37.5	36.7	30.7	35.5	28.3	27.1	32.1	30.5	9.8
15.9	20.1	20.4	27.7	26.7	3.9	-6.3	7.6	10.4	18.5	-13.2	-12.4
8.4	49.1	46.9	54.2	11.9	18.4	30.3	29.1	38.	18.3	34.6	34.5
36.	16.1	21.2	25.1	8.6	-0.3	1.8	15.4	46.4	50.6	34.3	43.
41.1	39.6	27.9	48.6	53.9	46.5	43.6	58.4	52.5	63.3	58.8	65.1
64.7	56.8	62.6	51.1	60.3	57.1	64.3	53.4	54.8	72.8	62.8	70.4
60.1	69.1	74.7	78.1	80.9	78.6	82.7	87.2	83.7	77.5	78.8	74.1
77.7	78.9	64.9	64.8	51.3	55.4	47.4	51.5	48.3	47.8	50.8	55.9
29.9	35.3	14.2	28.7	37.3	36.4	45.3	24.8	30.2	27.	33.1	47.6
14.	21.9	23.5	36.2	25.9	24.3	23.9	47.3	61.5	28.1	17.8	25.6
11.4	15.1	18.9	31.7	32.	49.4	19.3	37.4	18.1	43.2	12.5	14.4
12.9	45.6	49.7	35.9	43.7	52.6	41.2	44.3	47.5	38.1	52.8	51.4
51.8	48.7	60.6	56.3	61.8	61.1	81.5	81.8	79.3	79.4	70.5	82.5
80.7	85.8	80.2	78.7	78.5	79.1	75.2	75.	77.2	77.8	67.7	64.2
52.4	62.5	54.7	56.9	46.1	54.3	58.	46.7	48.2	27.3	47.7	35.1
28.4	57.9	36.3	30.1	23.1	50.1	28.6	24.6	29.2	33.7	39.8	4.2
4.6	22.5	19.5	32.7	24.4	17.6	21.7	10.3	17.4	13.	22.4	17.9
15.8	20.	13.5	40.4	25.5	44.7	55.1	47.	45.5	43.9	38.2	76.8
46.2	82.9	85.1	83.3	79.8	78.2	62.1	66.4	65.2	61.2	47.9	28.8
29.5	42.1	21.1	49.5	39.2	36.5	15.6	5.8	13.6	8.	7.9	9.3
10.6	11.7	6.4	8.3	14.7	6.7	18.2	-0.6	18.7	15.	8.5	16.8
8.2	40.7	51.2	36.6	17.3	12.3	34.2	58.6	51.6	60.9	65.5	83.9

80.4	80.3	79.9	62.3	32.5	37.7	52.9	45.1	42.5	54.9	31.5	32.6	
-6.9	-16.5	-1.5	17.7	41.7	2.7	10.7	39.4	37.	41.3	43.1	84.	
81.9	41.	45.	51.	11.6	5.1	-0.5	14.3	14.5	29.4	53.8]		
dew: [	28.3	38.9	42.6	23.4	11.5	3.5	1.1	-0.8	2.4	25.2	38.4	49.3
22.6	21.5	20.4	25.3	45.3	36.1	32.7	41.3	46.7	46.4	34.3	35.9	
31.4	22.2	20.3	18.3	15.3	25.7	27.6	16.7	9.	5.7	17.3	26.6	
47.9	36.7	14.8	9.3	30.9	39.9	18.5	21.7	23.2	21.6	39.2	35.6	
43.5	51.3	48.1	15.7	24.3	41.	47.7	21.9	13.5	8.5	6.7	34.2	
47.8	20.2	6.	9.1	12.8	17.4	7.8	13.3	18.4	33.3	29.6	30.	
32.1	10.6	5.	49.2	48.4	49.1	49.9	34.9	34.	47.1	36.3	39.8	
48.7	44.3	28.1	26.3	44.6	39.4	31.5	44.9	54.	41.9	32.6	49.8	
58.6	50.3	36.6	31.2	37.5	48.6	52.9	54.8	46.6	57.4	57.9	57.5	
39.	33.7	40.2	51.1	40.7	25.8	26.1	36.4	45.2	43.4	42.7	35.8	
56.1	61.3	54.6	59.	52.5	46.9	53.6	53.7	56.6	60.7	55.2	52.7	
52.2	47.3	43.7	44.8	54.9	59.7	48.8	55.3	66.1	66.4	67.2	65.6	
64.7	66.	61.4	52.8	57.6	65.3	45.5	65.7	65.9	60.	60.8	62.6	
63.	66.5	65.2	60.2	58.8	66.3	68.6	68.1	59.9	64.	64.4	67.1	
67.8	70.	69.5	63.1	55.6	58.5	63.9	54.3	57.	59.8	62.1	62.3	
63.6	60.3	53.5	63.2	60.6	56.	60.5	63.7	65.4	66.6	59.4	66.2	
50.6	57.1	58.9	49.4	51.	48.	48.3	45.	47.6	50.8	58.1	58.4	
58.3	61.2	62.5	62.2	61.	61.6	61.8	58.	59.5	62.4	63.5	61.7	
46.2	37.4	40.1	44.4	49.7	58.7	57.8	66.9	54.2	38.7	35.1	41.5	
49.	36.8	34.8	36.5	41.6	30.6	35.5	38.6	53.8	33.2	27.1	28.8	
13.7	8.9	22.5	33.9	28.4	40.	30.1	23.3	19.3	19.7	29.1	24.5	
27.4	29.2	24.1	32.5	29.5	29.3	19.8	11.9	13.1	18.7	9.6	15.5	
27.7	35.	23.	38.8	40.9	11.7	2.6	-0.7	5.3	9.8	1.	-1.1	
-1.4	-2.2	3.3	-7.8	-6.3	-3.8	22.7	28.6	8.3	0.5	17.6	1.5	
5.6	12.4	23.5	39.1	34.5	22.3	17.8	20.9	30.3	11.4	10.5	27.5	
5.4	7.5	28.7	11.3	22.	14.	21.2	32.9	19.9	16.2	53.4	36.2	
42.3	50.2	31.3	42.4	17.5	14.4	12.7	27.9	18.8	16.5	15.6	20.	
13.4	14.9	13.6	19.5	22.8	18.	11.	36.	52.	44.1	34.4	19.	
17.9	24.	25.1	37.1	35.4	26.9	31.9	22.1	23.7	37.8	39.3	21.	
29.8	59.6	47.5	51.7	50.4	54.7	54.4	61.5	64.2	62.9	62.8	59.2	
64.3	56.5	66.7	67.	67.5	69.1	64.8	60.4	65.5	59.3	49.6	68.5	
70.5	64.9	58.2	69.4	70.7	70.3	71.6	64.5	53.2	69.7	69.6	56.4	
62.	67.3	68.	68.4	64.1	69.	69.8	61.1	68.8	68.3	68.7	55.8	
67.9	71.2	70.4	70.1	54.5	68.2	65.1	53.1	50.7	52.3	60.9	65.	
56.9	67.7	63.8	34.1	33.1	32.	42.8	41.8	39.7	45.8	29.9	31.7	
31.	12.5	17.7	23.6	19.6	38.5	43.8	16.6	19.2	32.4	33.4	30.4	
20.7	26.8	29.4	29.	27.	28.2	15.	13.9	23.1	20.1	25.6	24.4	
31.1	-0.6	6.8	35.3	30.2	8.4	14.3	12.9	16.3	-7.1	-10.8	5.9	
37.9	45.4	46.8	53.3	7.2	37.	17.1	22.4	24.2	29.7	25.	34.6	
8.1	21.3	26.	10.8	4.4	28.9	40.3	24.9	18.6	40.8	19.1	18.9	
25.5	17.	27.8	41.1	44.2	48.9	52.6	37.2	48.2	34.7	55.4	50.	
51.4	51.2	56.3	57.7	50.5	42.5	40.6	55.9	53.	38.3	46.5	47.2	
55.1	51.9	66.8	69.9	70.8	72.7	74.5	73.	56.8	64.6	57.3	55.5	
49.5	47.	41.7	55.7	53.9	45.1	43.9	46.3	51.6	44.	51.8	26.7	
27.2	37.7	33.5	28.5	38.	28.	25.9	16.8	26.5	43.1	48.5	24.6	

15.1	37.3	14.1	16.1	13.	35.7	27.3	25.4	33.6	26.2	8.8	10.2
9.5	12.3	42.1	44.5	18.1	43.3	23.8	57.2	41.2	39.6	47.4	31.8
33.8	19.4	43.2	60.1	62.7	38.2	50.1	52.4	67.6	63.3	56.7	61.9
59.1	69.3	55.	40.4	45.9	37.6	20.8	45.7	32.8	30.5	21.4	33.
26.4	10.	20.6	31.6	9.4	23.9	10.3	12.	7.7	16.9	42.9	15.9
30.7	65.8	54.1	69.2	71.3	24.7	22.9	41.4	43.6	14.7	30.8	4.7
16.4	6.1	0.7	3.9	17.2	4.2	1.7	5.5	11.2	21.1	14.6	10.4
8.	39.5	9.7	42.	18.2	44.7	56.2	50.9	46.	8.6	21.8	38.1
32.3	5.2	-5.1	4.6	7.3	14.5	12.6	46.1	40.5	63.4	4.5	7.4
0.1	20.5	36.9	42.2	12.1	32.2]						
humidity: [71.1 87.1 91. 67.4 68.4 67.7 64.1 60.4 54.1 67. 70.1 75. 66.7 66.9											
67.5	74.1	84.3	80.2	81.5	85.	78.5	74.7	84.5	89.	74.	76.5 72.3 63.6
74.6	72.6	58.	57.1	49.3	45.3	80.4	77.8	72.7	60.3	54.6	55.6 54.4 65.6
69.1	57.5	43.5	62.9	63.5	44.9	57.7	72.9	55.5	54.	58.1	76.6 75.8 64.5
59.7	51.3	39.	63.3	82.	50.	34.7	53.6	57.4	51.9	70.4	61.5 56.5 63.4
87.5	62.5	42.5	35.2	49.	53.9	69.5	71.2	81.6	58.9	55.4	87.7 83.8 60.7
52.2	68.3	55.7	77.5	77.3	41.1	37.8	28.4	52.7	56.9	48.1	39.1 47.5 63.2
54.3	39.7	77.6	46.2	78.4	64.4	61.6	82.4	65.5	65.9	53.2	58.3 80.1 86.2
45.1	39.8	77.1	77.2	65.4	52.3	56.7	67.6	74.5	79.9	62.8	74.8 83.1 82.9
66.2	58.2	69.6	59.4	46.7	42.7	79.2	73.	75.4	68.2	56.1	55.3 71.4 80.6
80.8	82.3	72.8	71.9	76.7	73.8	87.3	67.2	50.9	60.	57.8	60.5 60.8 66.1
65.3	81.7	74.2	59.6	83.3	75.3	81.1	76.9	70.2	66.6	72.2	88. 86. 68.9
73.9	82.6	70.7	62.	62.1	68.	67.8	73.5	76.8	82.8	68.8	74.3 66.5 64.3
73.4	76.1	72.5	74.9	73.7	69.7	65.8	61.8	75.1	65.7	75.7	75.2 71. 83.7
63.8	64.2	61.7	79.3	85.4	65.2	80.9	84.8	76.	84.	71.3	72.4 62.7 56.8
50.1	59.8	78.2	74.4	77.9	57.3	91.3	73.2	69.9	84.6	53.1	47.8 54.9 79.8
64.9	56.	41.9	68.5	47.3	46.6	47.4	37.1	77.4	58.7	51.	57.9 54.7 70.6
68.7	63.1	72.	69.8	55.1	93.1	65.	55.	57.2	78.9	94.4	83.6 50.4 53.7
70.8	56.3	69.3	56.2	81.4	84.9	94.6	82.7	86.4	76.2	75.6	56.4 88.7 46.3
83.2	58.4	48.7	78.8	61.1	61.4	70.9	52.8	40.	41.2	90.1	58.6 44.5 53.8
83.4	99.2	96.	91.1	69.4	62.2	83.9	92.3	65.1	43.9	31.4	40.3 79.4 51.8
46.1	35.9	85.5	89.4	33.	30.	37.6	59.3	93.9	92.4	84.4	77. 82.2 70.
88.1	83.	71.5	85.2	84.2	85.7	71.7	91.2	88.8	52.9	85.1	83.5 80. 86.7
82.1	69.2	71.6	86.5	75.5	52.5	62.3	61.2	55.2	61.3	60.6	84.1 87.8 91.7
80.7	88.2	79.6	88.5	90.	81.8	86.8	81.3	73.3	69.	95.5	94.9 81.2 93.7
87.6	78.	78.1	80.3	85.6	96.8	89.6	89.1	92.2	90.5	92.8	93.5 70.3 66.
66.3	59.2	97.6	89.9	79.7	64.7	86.1	93.8	85.9	98.7	88.9	62.6 87.2 95.
77.7	81.	63.9	79.1	97.2	96.1	76.4	92.	87.	63.7	79.	86.6 93.2 89.3
78.3	61.9	95.9	66.4	53.3	64.6	42.8	53.4	59.5	86.3	67.9	52.6 42.4 41.6
45.5	52.1	43.3	53.5	39.4	40.8	35.1	37.4	38.2	45.	57.6	51.4 43.6 75.9
46.5	63.	67.1	89.2	91.9	85.3	90.7	60.9	82.5	59.9	64.	56.6 86.9 68.1
80.5	73.6	90.2	78.6	55.9	79.5	49.6	52.	58.8	55.8	68.6	95.2 90.4 91.8
92.6	97.7	92.9	91.6	60.2	39.5	32.7	92.1	48.5	60.1	41.8	29. 57. 90.9
67.3	48.	47.	81.9	93.3	44.4	47.2	42.9	35.7	51.7	50.6	59. 64.8 70.5
94.7	90.8	73.1	88.4	90.3	93.6	96.3	85.8	48.4	54.8	39.3	36.9 84.7 61.
92.5	91.5	66.8	94.3	87.9	89.5	88.6	88.3	45.4	44.1	37.5	34.2 42.6 38.3
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47.6	87.4	92.7	54.2	49.9	49.1	51.1	93.4	46.9	71.8	45.9	45.6 43.7 51.6

44.6 35.3 44.2 46.4 26. 32.8 24.7 37.7 28.7 31. 43. 43.1 54.5 47.9
 49.2 59.1 40.9 48.8 49.8 89.8 46. 48.2 50.2 35.4 95.6 36.8 36.4 40.4
 33.2 30.1 34.5 35. 30.7 32.2 38.6 49.5 35.8 36.7 42. 47.1 38.7 45.8
 76.3 91.4 36.6 72.1 50.8 41.7 42.1 46.8 48.3 37.9 45.7 52.4 31.3 51.5
 34. 40.7]

precip: [2.500e-02 9.500e-02 7.930e-01 5.200e-02 5.600e-02 8.000e-03 0.000e+00

1.080e-01 2.280e-01 4.840e-01 1.900e-02 8.300e-02 4.300e-02 1.930e-01
 3.000e-03 1.000e-02 9.720e-01 8.600e-02 5.500e-02 2.430e-01 1.800e-02
 1.500e-02 7.000e-03 2.300e-02 7.500e-02 4.800e-02 1.400e-02 4.640e-01
 2.140e-01 2.950e-01 1.000e-03 2.240e-01 1.100e-02 6.000e-03 3.230e-01
 1.600e-02 1.158e+00 6.300e-02 2.370e-01 5.120e-01 6.500e-02 3.860e-01
 1.530e-01 5.800e-02 2.340e-01 3.030e-01 1.510e-01 1.231e+00 6.800e-02
 2.400e-02 1.390e-01 5.340e-01 7.900e-02 1.270e-01 4.000e-03 3.700e-01
 3.270e-01 1.030e-01 9.350e-01 1.340e-01 4.400e-02 4.100e-02 1.230e-01
 1.900e-01 5.000e-03 4.070e-01 4.790e-01 8.900e-02 4.160e-01 4.480e-01
 2.600e-02 6.290e-01 3.600e-02 2.200e-02 1.049e+00 5.700e-02 5.790e-01
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 1.178e+00 1.090e+00 1.511e+00 8.570e-01 5.380e-01 6.980e-01 1.160e-01
 1.141e+00 2.100e-01 3.710e-01 3.120e-01 1.680e-01 8.740e-01 1.240e-01
 2.500e-01 3.800e-01 4.770e-01 1.190e-01 3.440e-01 1.860e-01 8.250e-01
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 3.100e-01 1.970e-01 1.070e-01 2.840e-01 3.700e-02 4.610e-01 4.700e-02
 5.300e-02 1.820e-01 5.540e-01 5.440e-01 3.970e-01 4.150e-01 4.900e-02
 1.100e+00 1.700e-02 4.030e-01 6.390e-01 2.260e-01 2.700e-02 9.080e-01
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 2.000e-03 3.100e-02 6.900e-01 2.000e-02 2.900e-02 2.890e-01 2.262e+00
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 3.400e-02 1.255e+00 2.470e-01 3.550e-01 4.830e-01 1.300e-01 3.810e-01
 6.220e-01 3.940e-01 4.330e-01 2.670e-01 1.940e-01 8.980e-01 3.900e-02
 8.000e-02 7.950e-01 7.440e-01 3.110e-01 5.530e-01 1.630e-01 3.000e-02
 1.640e-01 1.020e-01 2.710e-01 5.090e-01 2.930e-01 7.590e-01 2.110e-01
 2.830e-01 5.590e-01 6.860e-01 3.150e-01 2.335e+00 2.720e-01 4.200e-02
 2.290e+00 2.750e-01 4.520e-01 8.400e-02 6.240e-01 3.470e-01 5.100e-02
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 1.910e-01 6.690e-01 7.190e-01 1.040e+00 6.080e-01 4.980e-01 3.390e-01
 5.260e-01 2.580e-01 2.210e-01 2.060e-01 1.601e+00 3.922e+00 1.915e+00
 8.630e-01 1.610e-01 7.620e-01 3.480e-01 7.660e-01 1.440e-01 6.100e-02
 5.030e-01 1.161e+00 7.300e-02 2.560e-01 1.570e-01 8.800e-02 1.220e-01
 8.100e-02 3.280e-01 8.090e-01 3.350e-01 1.810e-01 8.200e-02 9.100e-02
 1.310e-01 3.170e-01 8.210e-01 1.960e-01 9.190e-01 7.200e-02 5.320e-01
 5.200e-01 2.010e-01 4.260e-01 1.620e-01 6.900e-02 7.560e-01 6.750e-01
 9.320e-01 7.360e-01 9.800e-02 3.800e-02 2.200e-01 6.250e-01 3.430e-01
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6.950e-01 6.890e-01 1.060e-01 1.690e-01 8.810e-01 2.540e-01 1.934e+00
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2.780e-01 5.190e-01 3.760e-01 4.860e-01 7.100e-02 8.050e-01 1.000e-01
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3.380e-01 4.560e-01 3.200e-02 4.720e-01 6.400e-02 1.730e-01 6.050e-01
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2.730e-01 4.110e-01 1.197e+00 8.030e-01 5.630e-01 1.880e-01 3.490e-01
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2.360e-01 8.180e-01 1.830e-01 9.730e-01 1.500e-01 2.660e-01 3.300e-01
1.010e-01 6.560e-01 2.313e+00 4.500e-02 6.710e-01 1.320e-01 3.310e-01
1.470e-01 1.750e-01 9.950e-01 7.130e-01 3.980e-01 2.770e-01 1.490e-01
6.210e-01 2.070e-01 7.320e-01 5.280e-01 2.460e-01 6.680e-01 4.000e-01
1.280e-01 3.400e-01 6.070e-01 8.220e-01 1.707e+00 2.860e-01 1.450e-01
8.730e-01 1.350e-01 6.700e-01 3.590e-01 9.380e-01 1.720e-01 1.660e-01
1.140e-01 1.210e+00 1.067e+00 5.900e-01 1.019e+00 1.137e+00 1.023e+00
1.220e+00 2.880e-01 7.500e-01 1.180e-01 3.610e-01 7.850e-01 1.139e+00
3.880e-01 1.202e+00 5.520e-01 9.030e-01 2.329e+00 1.770e-01 6.200e-01
1.700e-01 4.940e-01 5.580e-01 2.600e-01 1.330e-01 6.840e-01 1.053e+00
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4.990e-01 5.360e-01 8.260e-01 2.000e-01 2.040e-01 3.450e-01 2.160e-01
1.870e-01 5.750e-01 2.740e-01 3.990e-01 2.490e-01 5.430e-01 9.290e-01
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8.310e-01 2.120e-01 3.240e-01 6.960e-01 2.590e-01 2.850e-01 8.540e-01
6.320e-01 9.130e-01 4.230e-01 6.510e-01 5.900e-02 1.740e-01 1.348e+00
3.530e-01 4.490e-01 9.450e-01 3.540e-01 7.900e-01 2.330e-01 6.440e-01
6.790e-01 1.530e+00 1.464e+00 1.526e+00]
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precipcover: [ 12.5  29.17 70.83 25.      4.17  0.      37.5   8.33 20.83
79.17
16.67 33.33 41.67 45.83 62.5  91.67 75.      54.17 66.67 50.
58.33 83.33 87.5  95.83 100.  ]
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4.1 1.3 2.2 0.7 2.1 2.7]
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1.2

1.9]

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29.5 24.5 32.9 21. 32.4 36. 23. 22.1 46.6 37.7 30. 13.4 30.8 15.9

23.5 27.6 38.8 28.4 34.4 46.8 38.2 33. 19.7 17.4 24.2 34.6 46. 31.3

32.6 19.2 33.4 35.7 27. 21.4 16.6 18.3 24.7 20.8 28.7 25.1 38.5 28.2

29.1 11.6 24.4 33.8 32.3 20.4 30.6 33.3 18.8 11.9 23.8 35.3 15. 26.8

25.3 16.8 23.7 29.8 56.4 45.4 27.3 27.2 25.8 13.9 12.8 27.8 30.1 17.

27.1 21.6 23.1 25. 26.4 21.5 23.9 40.3 29.9 41.4 18.1 39.8 27.9 27.7

26.3 20.6 16.3 25.7 15.7 36.7 22.6 19.5 11. 16.1 32.1 9.6 35.6 22.4

10.7 20.1 32.2 10.5 28.1 9.4 9.8 14.1 29.7 15.2 12.3 24. 27.5 38.

13.2 35.4 14.8 12.1 18.6 47.6 28.8 26.1 34. 37.5 15.4 17.7 33.5 31.6

51.2 41.7 37.8 26.7 22.2 7.8 30.3 19.1 25.4 34.2 26.5 14.3 22.7 33.2

31.2 39.4 24.9 26. 29.4 35.1 38.6 39.2 38.7 34.3 39. 32. 39.1 34.9

33.6 45.2 34.8 26.2 39.9 32.7 36.6 47.8 44.9 31.7 31.4 30.9 50.8 26.6

44.3 32.8 33.7 10.3 11.2 34.1 36.8 23.6 12.5 11.4 45.5 14.5 37.6 42.1

44.4 20.5 26.9 7.6 30.4 24.8 41.6 42.7 45.6 45.7 37. 38.9 29.6 39.5

43.9 40.2 38.3 56.1 49.6 37.9 36.5 37.3 19.8 40.8 36.3 33.1 36.1 37.4

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40.7 47.2 35. 25.6 37.1 36.4 50.1 36.9 34.5 40. 45.1 47.3 44.8 39.7

41. 48.6 7.2 8.9 52.1 41.3 57.9 50.6 8.5 8.1 6. 16.4 39.3 7.4

46.9 5.8 50.9 20.3 45. 43.3 49.5 61.2 18.4 19.4 31.9 21.1 41.2 52.5

16.9 40.6 43.4 22.9 47.9]

windspeed: [15.4 9. 10.3 22.5 13.8 10.2 13.2 14.8 12.3 23.7 24.1 26.4 10.7

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14.9 17.4 14. 11.7 19.4 20.7 15.9 22.3 13.6 12.5 25.4 21.9 17.2 18.7

13.7 8.3 13.3 8.4 8.9 15.2 16.7 21.3 18.4 23.9 20.8 11.5 9.1 14.5

19.9 28. 20.2 16.1 9.9 20.9 22.9 11.9 15.5 18.9 12.9 16.8 14.3 10.

15.1 13.4 14.6 7.1 27. 20.6 11.3 18.2 17.5 10.9 5.7 23.4 11.6 14.2

9.4 16.6 10.4 18. 23.6 19.3 7.9 21.1 10.5 11.2 11.4 12.8 5. 18.8

13.9 12.1 9.6 7. 16.2 9.8 19. 15.8 16.3 16.4 13.5 10.6 8.7 12.2

11. 17.6 8. 6.5 11.1 12.7 8.6 7.3 6.1 14.7 6.2 6. 9.2 8.1

6.6 17.7 6.4 6.3 7.4 9.5 4.9 5.3 4.6 7.5 5.1 6.7 16.5 8.8

14.1 5.9 8.2 22.2 23.3 7.7 9.3 19.7 19.1 21.5 10.8 20.1 19.2 16.

17.3 20.3 18.1 7.8 17.9 16.9 12. 28.9 26.2 10.1 17.1 28.3 18.3 25.8

19.6 22.7 22. 25.5 7.2 15.7 5.5 6.8 6.9 8.5 13.1 15.6 4.8 22.6

5.2 21.8 18.5 25.9 23.1 24.2 30. 34.8 24.4 24.8 22.8 24.7 5.4 21.

21.4 31.3 22.1 25.1 19.8 17.8 25.6 21.2 23. 4.3 21.7 23.5 24. 24.5

28.5 27.8 21.6 27.9 28.2 22.4 23.8 25. 29.6 4.4 23.2 20.5 29.5 28.1

27.1 29.2 4.1 24.9 20.4]

winddir: [224. 112.6 127.3 ... 104. 155.1 262.8]

sealevelpressure: [1021.5 1021.8 1006.3 1007.1 1015.1 1022.1 1029.3 1035.6

1035.3 1020.9

```

1019.1 1015.6 1036. 1031.8 1030. 1027.2 1014.9 1017.5 1018.5 1009.1
1008.2 1002.9 1000.9 1005.8 1003.5 1010.2 1007.9 1008. 1012.4 1007.6
1016.1 1022.5 1026.5 1016.2 1018.1 1004.1 1006. 1023.7 1014.1 1009.9
1011.3 1003.7 1011.9 1012.2 1009.5 1015.4 1023.8 1012.8 1009.8 1006.7
1004.6 1020.6 1026.3 1020. 1017.6 1037. 1034.6 1025.7 1017. 1017.9
1018.9 1027.6 1029.8 1024.3 1012.3 1016.9 1025.1 1017.3 1025.2 1021.4
1015.8 1028.1 1033.2 1022.3 1018.2 1014.3 1013.2 1014.8 1004.9 1023.5
1009.7 1002. 995.5 1006.1 1017.4 1019. 1020.4 1025.9 1029.6 1021.3
1025. 1021.1 1016. 1017.1 1018.7 1016.7 1005.5 1018.4 1015.7 1005.
1018.8 1015.3 998.5 999. 1008.8 1015.5 1015.2 1011.4 1012.9 1011.
1016.6 1016.8 1013.9 1020.8 1016.5 1003.1 1008.7 1012.7 1011.7 1017.8
1010.7 1015.9 1019.4 1018.6 1012.5 1010.4 1007.7 1010. 1017.2 1014.5
1016.4 1016.3 1015. 1008.9 1011.5 1021. 1013.8 1019.2 1019.5 1018.3
1013.3 1019.6 1023.4 1022. 1010.6 1013.6 1020.1 1014. 1024.9 1021.2
1030.2 1025.5 1010.9 1013.7 1019.9 1021.6 1020.3 1020.2 1018. 1019.8
1026.6 1028. 1031.5 1022.4 1019.7 1010.8 1023.2 1027. 1023. 1024.6
1024.4 1007.3 1013.5 1010.5 1006.8 1021.7 1023.3 1024.8 1032.9 1027.9
1028.4 1001.2 1020.5 1020.7 1014.6 1009. 1022.6 1024.1 1023.6 1023.1
1017.7 1006.2 1011.1 1022.2 1036.1 1037.2 1026.9 1029.9 1033.9 1034.5
1022.7 1033. 1032.3 1025.6 1024.7 1007.2 1038.4 1030.7 1029.7 1032.8
1032.4 1026.1 1030.8 1027.3 1033.5 1037.5 1026.2 1032.6 1025.4 1012.6
1027.1 1029. 1010.3 1011.8 1007.5 1024. 1023.9 1028.3 1011.6 1008.1
1004.3 1026.8 1013.1 1014.2 1019.3 1012.1 1012. 1026.7 1008.3 1014.7
1013.4 1021.9 1022.8 1014.4 1024.2 1022.9 1007. 1030.9 1025.8 1005.1
1026. 1032.7 1001.3 1009.4 992.5 1011.2 1028.6 1032.2 1006.4 1013.
1006.5 1030.3 1042. 1033.7 1002.3 1030.5 1025.3 1030.1 1010.1 1028.8
1004.2 1008.5 1000.4 1000.2 1009.6 1007.8 1008.4 1008.6 1002.4 1009.2
1005.9 1028.9 1027.8 1007.4 1000.3 1003. 1035.1 1001.4 1034.8 1031.6
1002.5 1033.8 1040.1 1035.2 1031.2 998.1 1030.4 1029.2 1031. 1031.3
999.6 1009.3 1005.2 1027.5 1027.4 1034.3 1026.4 1003.4 1006.6 1028.2
1031.9 1034.4 1031.4 999.3 1028.5 1000.1 1029.4 1002.8 1005.6 1031.7
997. 1032. 997.7 1004.5 1024.5 1036.7 1027.7 1002.7 1005.4 1029.1
1004.7 1029.5 1004.8 1003.3 994.2 997.6 1002.1 1006.9 1036.6 1038.9
1001.8 997.4 999.1 1035.8 1033.3 1001.5 998.6 1001.6 996.7]
cloudcover: [ 41.7 87.9 78.7 73. 83.5 31.8 58. 46.5 58.8 80.3 41.6
86.6
58.1 77.7 28.7 71.9 82.6 93.8 75.4 69.4 79.1 90.9 96.6 80.1
81.3 82.3 95.6 84.2 72.3 84.9 50.8 49.8 10.6 91. 18.7 83.8
88.5 71.7 33.8 85.7 52.4 26.4 80.8 70. 33.6 16.2 26.9 28.8
32.8 87.1 74.2 56. 80. 67.4 70.2 84.4 63.8 13.1 13.6 90.7
82.1 22.5 41.9 75.5 55.7 59. 35.4 87.2 61.7 39.8 94. 97.4
84.7 69. 35. 6.9 39.9 41.1 51.8 54. 54.7 55.4 84.5 98.3
26.8 49.1 69.9 36.8 81. 76.5 5.8 16.1 20.1 47.4 36.5 36.6
61.4 54.9 33.7 13.9 51.4 54.8 11.5 50.3 70.1 37.1 27.8 45.3
44.7 57.6 82.7 64.4 68.2 77.5 87.6 71.1 45.8 44.1 85.5 97.8
60.3 12.8 9.3 15.7 43.9 64.5 55.2 86.5 77.6 70.5 67.1 39.7
30. 16.6 21.6 65.4 40.1 51.7 6.3 28.9 35.5 38.8 38.1 48.5
24.2 32.5 64.3 25.6 35.7 38.2 43. 46.4 48.4 9.6 29.8 47.

```

18.1	25.2	36.3	29.9	54.1	60.2	62.4	38.4	58.7	47.5	53.9	52.8
30.8	25.8	16.7	32.1	76.9	82.	71.2	78.8	12.	33.5	49.2	27.1
8.5	17.8	31.7	23.9	31.	53.1	29.1	6.5	50.6	57.9	37.7	21.8
43.1	44.9	30.9	27.6	54.2	27.7	47.7	21.4	37.5	39.4	20.7	80.2
52.1	48.8	13.	5.	27.2	57.4	72.2	73.6	56.4	40.4	35.9	12.4
26.3	24.	14.	7.5	5.5	17.1	19.2	31.9	22.2	6.2	13.2	72.6
79.6	29.2	84.	44.2	73.8	82.4	46.1	12.7	42.1	51.2	2.9	4.9
13.3	10.9	16.3	56.2	66.3	69.2	53.5	75.1	93.1	40.6	70.8	60.6
48.3	67.3	72.8	21.2	26.5	65.6	46.3	89.1	80.6	37.2	78.2	80.9
33.2	14.7	30.7	4.	52.7	40.9	7.8	32.9	27.3	19.3	11.9	75.
34.2	51.5	90.	79.4	77.3	70.9	59.2	56.6	90.6	16.8	49.7	93.3
88.8	65.1	63.1	50.1	36.7	91.7	78.6	48.1	18.6	43.5	42.6	99.1
53.8	77.1	99.5	84.8	20.4	98.2	55.3	5.4	7.	14.9	99.9	87.8
99.7	85.4	64.2	41.3	56.5	89.7	36.1	95.7	96.4	74.6	99.8	99.3
99.	79.9	69.5	91.8	100.	53.3	69.3	36.	93.	62.6	4.7	1.7
9.5	81.6	95.5	91.9	24.4	9.9	78.5	83.3	88.	63.7	93.9	38.
15.9	7.6	11.7	99.4	89.5	42.5	76.7	92.3	67.2	88.7	53.	74.1
53.2	82.8	2.6	16.5	19.8	35.2	70.3	65.5	0.	3.3	77.8	24.6
90.4	68.7	68.8	5.9	20.2	74.7	40.8	66.4	75.8	43.7	62.8	42.2
68.	90.5	83.	42.9	87.4	41.4	52.9	89.8	81.4	65.	59.3	63.9
20.5	11.4	9.	44.5	30.4	56.3	4.3	11.3	35.8	22.6	11.2	17.2
19.6	27.	25.4	19.5	36.2	63.5	52.5	43.6	39.1	72.9	34.7	37.6
65.8	74.4	59.5	23.3	77.9	63.2	73.4	65.3	86.	27.9	55.1	15.2
35.6	68.6	19.4	68.3	16.	82.2	34.1	21.1	13.8	56.1	95.1	92.1
28.4	86.3	78.1	26.	51.3	55.9	68.9	94.6	81.8	35.3	26.1	22.1
78.3	61.2	91.5	62.9	92.7	96.7	22.8	50.2	94.2	42.4	57.	30.6
70.7	93.2	94.8	76.8	19.	66.9	12.5	24.8	80.4	15.	45.1	84.3
97.	96.	63.4	23.1	72.4	47.2	65.9	88.4	57.1	13.7	79.	98.8
44.6	90.8	66.6	67.7	90.1	95.2	30.1	45.4	75.2	62.1	78.4	85.
68.1	89.	92.5	77.	60.7	71.8	95.4	45.6	33.3	80.7	87.	52.2
86.8	69.8	33.1	82.9	22.9	61.1	85.6	78.	74.3	90.3	86.7	60.5
59.4	62.5	29.3	83.1	25.7	10.8	86.1	8.	68.5	31.4	49.5	61.5
64.	45.	50.	58.6	45.7	48.2	81.2	79.2	69.7	89.2	14.6	20.9
43.4	94.1	84.6	64.9	64.7	67.6	57.8	36.4	53.7	23.	45.5	45.9
61.3	58.2	56.7	43.2	57.7	15.8	72.1	12.9	48.6	73.2	55.	18.8
59.8	34.3	38.3	60.1	32.4	51.9	4.1	8.7	57.2	29.7	44.8	42.
47.9	25.9	11.1	92.	67.8	30.2	24.9	25.3	51.	7.7	46.	38.7
10.4	27.5	33.4	17.4	17.6	21.5	63.6	24.1	7.3	15.6	47.1	17.9
55.5	54.4	74.5	12.2	6.6	12.3	70.4	85.9	10.2	15.4	66.	44.3
62.3	25.1	43.8	71.3	39.2	10.5	20.3	67.5	97.3	97.1	81.5	14.3
19.1	90.2	88.2	65.7	50.7	68.4	91.6	52.6	49.3	56.8	15.1	95.9
75.3	79.5	58.9	92.8	59.1	18.	26.6	1.4	1.2	49.6	84.1	76.6
72.7	49.9	9.7	21.	66.1	76.2	24.7	79.7	3.1	4.2	66.5	59.7
64.8	37.9	20.	54.5	85.8	87.5	83.9	41.5	60.	73.7	32.3	73.1
7.2	47.8	59.9	22.	23.5	63.3	9.4	49.	5.6	55.6	50.5	18.4
42.3	44.	32.6	18.3	14.8	31.3	67.9	6.1	37.	20.6	46.6	55.8
70.6	48.	47.6	37.4	46.8	23.6	30.5	11.	6.	73.3	37.3	28.5
14.1	14.4	5.2	0.6	7.9	10.	78.9	76.3	62.7	26.7	38.6	10.3

```

53.4 12.6 14.2 81.7 99.6 13.4 19.9 8.3 88.3 93.7 96.3 96.8
17.5 96.1 28.6 95.3 39.6 98.1 77.2 94.7 94.3 61. 71.6 34.9
42.8 98.9 92.6 99.2 91.4 65.2 37.8 74.8 31.1 18.2 0.3 21.3
41.8 31.6 40.5 46.2 40.2 74.9 38.9 28.1 24.5 41. 76. 32.7
66.7 69.1 34.4 9.1 85.1 23.2 20.8 51.6 86.4 40.7 24.3 35.1
69.6 91.2 79.8 7.4 56.9 29.4 91.1 4.4 97.2 15.5 22.7 85.3
89.4 3.2 49.4 96.2 80.5 97.6 46.9 3.5 2.1 2. 97.9 89.3
92.4 48.9 9.8 23.4 50.4 71.5 34. 5.3 75.6 33.9 3.8 96.9
54.6 72.5 62.2 98.7 59.6 97.7 93.6 76.4 17. 32.2 79.3 71.4
88.1 52. 29. 6.8 12.1 15.3 52.3 60.4 83.2 29.6 98.6 40.3
8.4 94.5 83.7 96.5 67. 21.7 19.7 88.9 50.9 61.6 94.4 18.9
5.1 6.7 81.1 86.9 18.5 66.8 11.6 17.7 47.3 85.2 62. 23.7
44.4 89.9 95.8 95. 92.2 3.6 22.4 14.5 30.3 3.9 2.2 31.2
76.1 25. 63. 36.9 28. 73.9 60.8 5.7 81.9 98. 38.5 11.8
28.3 93.4 2.7 92.9 87.7 6.4 22.3 17.3 39.5 10.1 66.2]
visibility: [9.5 5.5 5.3 9. 6.4 8.9 9.6 9.4 9.9 8.2 9.3 9.8 9.1 6.9 8.7 7.2 7.1
7.5
9.7 8.6 7.6 8.5 6.3 7.4 7.9 7.3 9.2 8.4 8.3 5.9 5. 7.7 4.5 6.2 6.6 8.1
8.8 8. 6.1 6.8 7. 6.7 3.6 6.5 4.3 4. 4.9 7.8 3.3 2.8 5.6 3.2 4.6 5.8
4.1 5.4 2.2 6. 4.4 1.9 5.7 5.2 3.1 4.2 1.1 3.7 4.7 3.9 2. 4.8 2.7 2.9
1.8 2.6 1.4 3.5 1.2]
solarradiation: [ 95.6 15.5 7.9 ... 256.7 177. 278.9]
solarenergy: [ 8.2 1.3 0.7 6.8 6. 9.1 8.4 8.3 8.1 1.4 5.8 9.4 3.3
2.5
2.1 6.5 1.5 7.3 1.1 3.9 3.8 7.8 6.4 7.7 10.2 4.2 8.7 12.2
13. 3.5 5.4 9.7 9. 10.1 4.3 11. 14. 11.5 14.6 14.3 14.2 8.6
5.9 8.8 12.6 12.7 7.5 8.9 4.4 10.4 16.2 17.7 17.8 5. 18.8 18.5
17.2 17.5 14.7 15.1 12.8 12.1 6.6 10.9 16.8 22.7 22.5 18.4 9.6 11.3
13.8 21.3 13.6 2.6 9.2 21.9 11.2 11.9 21.5 7. 25.2 24.7 23. 14.4
24.8 15.3 16.5 20.1 19.6 23.7 26.4 11.8 22.3 13.7 25.3 19.4 22.2 21.6
6.3 21.4 8.5 16.1 18.9 14.8 7.2 23.5 28.5 25.9 15.5 21.7 29.5 28.2
25.7 24.5 15.8 22.1 24.9 28.8 29.3 15. 18.2 10. 20.5 23.4 29.8 25.5
22.9 19.7 6.2 26. 24.1 29.2 18.7 20.3 25. 27.3 21.8 19.2 23.9 20.
19. 24.3 23.3 23.6 22.6 20.9 15.9 17. 23.1 15.4 27. 27.4 17.1 18.1
21.2 23.8 19.1 18.3 21.1 16.4 19.3 12.3 10.3 6.9 17.6 16. 8. 13.2
16.6 16.3 17.3 19.5 14.5 5.2 15.2 12.4 5.6 10.6 13.5 1.8 5.5 3.
1.9 4.7 3.7 9.3 10.7 9.9 4.5 9.8 7.1 7.4 9.5 7.9 0.9 6.1
6.7 4. 2.4 7.6 5.3 3.4 2.9 2.7 0.5 3.1 1.6 3.2 2.2 2.3
1.2 4.1 5.1 5.7 2.8 4.8 4.9 14.1 11.7 10.8 12.5 11.1 14.9 12.
10.5 2. 0.8 3.6 0.6 1. 1.7 11.4 4.6 15.7 19.8 15.6 16.7 13.4
18.6 20.6 12.9 17.4 18. 19.9 22.4 20.8 16.9 13.9 13.3 13.1 22. 26.7
17.9 25.4 25.1 24. 23.2 22.8 11.6 24.2 20.2 25.6 24.6 27.9 27.8 26.5
20.7 20.4 21. 26.1 24.4 26.6 26.8 26.9 26.2 27.2 26.3 25.8]
uvindex: [ 4 1 0 3 5 2 6 7 8 9 10]
severerisk: [ nan 10. 60. 30. 75. 100.]
sunrise: ['2017-01-01T07:43:20' '2017-01-02T07:43:25' '2017-01-03T07:43:27' ...
'2024-04-29T06:21:04' '2024-04-30T06:19:47' '2024-05-01T06:18:31']
sunset: ['2017-01-01T17:04:23' '2017-01-02T17:05:15' '2017-01-03T17:06:08' ...

```

```
'2024-04-29T20:14:17' '2024-04-30T20:15:18' '2024-05-01T20:16:20']
moonphase: [0.12 0.16 0.19 0.23 0.25 0.3 0.33 0.37 0.4 0.44 0.47 0.5 0.54
0.57
0.6 0.64 0.67 0.7 0.75 0.76 0.8 0.83 0.86 0.89 0.93 0.96 0. 0.03
0.06 0.1 0.13 0.17 0.2 0.27 0.31 0.35 0.38 0.42 0.45 0.52 0.55 0.59
0.62 0.65 0.68 0.71 0.78 0.81 0.84 0.87 0.91 0.94 0.97 0.04 0.07 0.11
0.15 0.18 0.22 0.29 0.32 0.36 0.43 0.53 0.63 0.66 0.73 0.79 0.82 0.92
0.95 0.02 0.05 0.09 0.34 0.41 0.48 0.58 0.61 0.74 0.14 0.28 0.39 0.46
0.56 0.69 0.85 0.88 0.9 0.98 0.08 0.21 0.24 0.77 0.72 0.26]
```

```
[ ]:
```

```
[ ]:
```

4.7 OverDoseDataPA1

```
[ ]: overdose_PA_df = pd.read_excel('../data/raw/OverdoseDataPA1.xlsx',
engine='calamine')
overdose_PA_df.head(5)
```

```
[ ]:   dose_count  incident_id incident_time      day incident_county_name \
0          1         184      13:52:00  Wednesday      Allegheny
1          2        39237       4:13:00   Saturday      Allegheny
2          1        27008       0:15:00    Monday    Lackawanna
3          2        47495      21:00:00   Saturday        Adams
4          1        25677      17:30:00   Wednesday      Allegheny

   incident_state  victim_id  gender_desc  age_range  race  ... \
0  Pennsylvania      124      Female    20 - 24  WHITE  ...
1  Pennsylvania     31680      Female    50 - 59  WHITE  ...
2  Pennsylvania     21028      Female    30 - 39  BLACK  ...
3  Pennsylvania     38590       Male    40 - 49  WHITE  ...
4  Pennsylvania     20086       Male    50 - 59  WHITE  ...

      geocoded_column  geocoded_column_address  geocoded_column_city \
0  POINT (-79.986198 40.467355)              NaN              NaN
1  POINT (-79.986198 40.467355)              NaN              NaN
2  POINT (-75.612183 41.439101)              NaN              NaN
3  POINT (-77.222243 39.872096)              NaN              NaN
4  POINT (-79.986198 40.467355)              NaN              NaN

   geocoded_column_zip  geocoded_column_1_city  geocoded_column_1_zip \
0              NaN              NaN              NaN
1              NaN              NaN              NaN
2              NaN              NaN              NaN
3              NaN              NaN              NaN
4              NaN              NaN              NaN
```

	geocoded_column_1_state	geocoded_column_1	dose_unit	\
0	NaN	POINT (-79.986198 40.467355)	0	
1	NaN	POINT (-79.986198 40.467355)	0	
2	NaN	POINT (-75.032709 41.332572)	0	
3	NaN	POINT (0 0)	2	
4	NaN	POINT (-79.986198 40.467355)	0	

	geocoded_column_1_address
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

[5 rows x 42 columns]

```
[ ]: overdose_PA_df
```

[]:	dose_count	incident_id	incident_time	day	incident_county_name	\
0	1	184	13:52:00	Wednesday	Allegheny	
1	2	39237	4:13:00	Saturday	Allegheny	
2	1	27008	0:15:00	Monday	Lackawanna	
3	2	47495	21:00:00	Saturday	Adams	
4	1	25677	17:30:00	Wednesday	Allegheny	
...	...	...	...	...	...	
53932	2	39326	14:27:00	Tuesday	Cambria	
53933	2	14560	16:36:00	Friday	York	
53934	1	12293	21:30:00	Monday	Allegheny	
53935	2	35341	21:58:00	Sunday	Westmoreland	
53936	0	25135	22:00:00	Friday	Fayette	

	incident_state	victim_id	gender_desc	age_range	race	...	\
0	Pennsylvania	124	Female	20 - 24	WHITE	...	
1	Pennsylvania	31680	Female	50 - 59	WHITE	...	
2	Pennsylvania	21028	Female	30 - 39	BLACK	...	
3	Pennsylvania	38590	Male	40 - 49	WHITE	...	
4	Pennsylvania	20086	Male	50 - 59	WHITE	...	
...	...	...	...	...	...	...	
53932	Pennsylvania	31751	Male	40 - 49	BLACK	...	
53933	Pennsylvania	11654	Male	25 - 29	WHITE	...	
53934	Pennsylvania	9750	Female	30 - 39	WHITE	...	
53935	Pennsylvania	28171	Male	40 - 49	WHITE	...	
53936	Pennsylvania	19712	Male	40 - 49	WHITE	...	

	geocoded_column	geocoded_column_address	\
0	POINT (-79.986198 40.467355)	NaN	
1	POINT (-79.986198 40.467355)	NaN	

2	POINT (-75.612183 41.439101)	NaN
3	POINT (-77.222243 39.872096)	NaN
4	POINT (-79.986198 40.467355)	NaN
...	...	...
53932	POINT (-78.718942 40.491275)	NaN
53933	POINT (-76.725761 39.921925)	NaN
53934	POINT (-79.986198 40.467355)	NaN
53935	POINT (-79.471341 40.310315)	NaN
53936	POINT (-79.651896 39.919448)	NaN

	geocoded_column_city	geocoded_column_zip	geocoded_column_1_city	\
0	NaN	NaN	NaN	
1	NaN	NaN	NaN	
2	NaN	NaN	NaN	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	
...	...	...	...	
53932	NaN	NaN	NaN	
53933	NaN	NaN	NaN	
53934	NaN	NaN	NaN	
53935	NaN	NaN	NaN	
53936	NaN	NaN	NaN	

	geocoded_column_1_zip	geocoded_column_1_state	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	
...	...	...	
53932	NaN	NaN	
53933	NaN	NaN	
53934	NaN	NaN	
53935	NaN	NaN	
53936	NaN	NaN	

	geocoded_column_1	dose_unit	geocoded_column_1_address
0	POINT (-79.986198 40.467355)	0	NaN
1	POINT (-79.986198 40.467355)	0	NaN
2	POINT (-75.032709 41.332572)	0	NaN
3	POINT (0 0)	2	NaN
4	POINT (-79.986198 40.467355)	0	NaN
...	...	...	...
53932	POINT (-78.718942 40.491275)	2	NaN
53933	POINT (-76.725761 39.921925)	4	NaN
53934	POINT (-79.986198 40.467355)	0	NaN
53935	POINT (-79.471341 40.310315)	2	NaN

```
53936 POINT (-79.651896 39.919448) 0 NaN
```

```
[53937 rows x 42 columns]
```

```
[ ]: overdose_PA_df.columns
```

```
[ ]: Index(['dose_count', 'incident_id', 'incident_time', 'day',
          'incident_county_name', 'incident_state', 'victim_id', 'gender_desc',
          'age_range', 'race', 'ethnicity_desc', 'victim_state', 'victim_county',
          'accidental_exposure', 'victim_od_drug_id', 'susp_od_drug_desc',
          'naloxone_administered', 'administration_id', 'incident_date',
          'dose_desc', 'response_time_desc', 'survive', 'response_desc',
          'revive_action_desc', 'third_party_admin_desc',
          'incident_county_fips_code', 'incident_county_lat',
          'incident_county_long', 'victim_county_fips_code', 'victim_county_lat',
          'victim_county_long', 'geocoded_column_state', 'geocoded_column',
          'geocoded_column_address', 'geocoded_column_city',
          'geocoded_column_zip', 'geocoded_column_1_city',
          'geocoded_column_1_zip', 'geocoded_column_1_state', 'geocoded_column_1',
          'dose_unit', 'geocoded_column_1_address'],
          dtype='object')
```

```
[ ]: overdose_sel_cols = [
    'incident_time', 'day', 'incident_county_name', 'gender_desc', 'age_range',
    'race', 'ethnicity_desc', 'accidental_exposure', 'susp_od_drug_desc',
    'naloxone_administered', 'incident_date', 'response_time_desc', 'survive',
    'incident_county_lat', 'incident_county_long']
```

```
[ ]: overdose_PA_df[overdose_sel_cols].describe()
```

```
[ ]:
      incident_county_lat  incident_county_long
count          53937.000000          53937.000000
mean             40.551382            -77.434264
std              0.566338             1.892110
min             39.854804            -80.351074
25%             40.167598            -79.762866
50%             40.419746            -76.725761
75%             40.815095            -75.711070
max             41.994138            -75.032709
```

```
[ ]: overdose_PA_df['day'].value_counts()
```

```
[ ]: day
      Saturday      8724
      Friday       8474
```

```

Thursday      7670
Wednesday     7453
Tuesday       7402
Sunday        7262
Monday        6952
Name: count, dtype: int64

```

```
[ ]: overdose_PA_df['incident_county_name'].count()
```

```
[ ]: 53937
```

```
[ ]: down_sel = overdose_PA_df[overdose_PA_df['incident_county_name'] == 'Allegheny']
down_sel
```

```
[ ]:
      dose_count  incident_id incident_time      day incident_county_name \
0              1          184    13:52:00  Wednesday      Allegheny
1              2         39237     4:13:00   Saturday      Allegheny
4              1         25677    17:30:00  Wednesday      Allegheny
6              1          3984     0:02:00   Friday      Allegheny
22             0         42561     8:00:00   Monday      Allegheny
...
53897          2         26867     0:00:00   Saturday      Allegheny
53904          2         39762    23:00:00   Monday      Allegheny
53918          0         23593     1:00:00   Saturday      Allegheny
53929          1         39064    21:16:00   Sunday      Allegheny
53934          1         12293    21:30:00   Monday      Allegheny
```

```

      incident_state  victim_id  gender_desc  age_range  race  ... \
0      Pennsylvania      124      Female    20 - 24  WHITE  ...
1      Pennsylvania    31680      Female    50 - 59  WHITE  ...
4      Pennsylvania    20086       Male    50 - 59  WHITE  ...
6      Pennsylvania     3252       Male    50 - 59  WHITE  ...
22     Pennsylvania    34747       Male    40 - 49  WHITE  ...
...
53897  Pennsylvania    20932      Female    20 - 24  WHITE  ...
53904  Pennsylvania    32155       Male    30 - 39  WHITE  ...
53918  Pennsylvania    18642       Male    25 - 29  WHITE  ...
53929  Pennsylvania    31523       Male    40 - 49  WHITE  ...
53934  Pennsylvania     9750      Female    30 - 39  WHITE  ...
```

```

      geocoded_column  geocoded_column_address \
0      POINT (-79.986198 40.467355)      NaN
1      POINT (-79.986198 40.467355)      NaN
4      POINT (-79.986198 40.467355)      NaN
6      POINT (-79.986198 40.467355)      NaN
22     POINT (-79.986198 40.467355)      NaN
...

```

53897	POINT (-79.986198 40.467355)	NaN
53904	POINT (-79.986198 40.467355)	NaN
53918	POINT (-79.986198 40.467355)	NaN
53929	POINT (-79.986198 40.467355)	NaN
53934	POINT (-79.986198 40.467355)	NaN

	geocoded_column_city	geocoded_column_zip	geocoded_column_1_city \
0	NaN	NaN	NaN
1	NaN	NaN	NaN
4	NaN	NaN	NaN
6	NaN	NaN	NaN
22	NaN	NaN	NaN
...	...	...	...
53897	NaN	NaN	NaN
53904	NaN	NaN	NaN
53918	NaN	NaN	NaN
53929	NaN	NaN	NaN
53934	NaN	NaN	NaN

	geocoded_column_1_zip	geocoded_column_1_state \
0	NaN	NaN
1	NaN	NaN
4	NaN	NaN
6	NaN	NaN
22	NaN	NaN
...	...	...
53897	NaN	NaN
53904	NaN	NaN
53918	NaN	NaN
53929	NaN	NaN
53934	NaN	NaN

	geocoded_column_1	dose_unit	geocoded_column_1_address
0	POINT (-79.986198 40.467355)	0	NaN
1	POINT (-79.986198 40.467355)	0	NaN
4	POINT (-79.986198 40.467355)	0	NaN
6	POINT (-79.986198 40.467355)	0	NaN
22	POINT (-79.986198 40.467355)	0	NaN
...	...	...	...
53897	POINT (0 0)	4	NaN
53904	POINT (-79.986198 40.467355)	4	NaN
53918	POINT (-79.986198 40.467355)	0	NaN
53929	POINT (-79.986198 40.467355)	4	NaN
53934	POINT (-79.986198 40.467355)	0	NaN

[7204 rows x 42 columns]

```
[ ]: down_sel['susp_od_drug_desc'].unique()
```

```
[ ]: array(['HEROIN', 'FENTANYL', 'PHARMACEUTICAL OTHER', 'UNKNOWN',
          'MARIJUANA', 'CARFENTANIL',
          'FENTANYL ANALOG/OTHER SYNTHETIC OPIOID', 'METHADONE', 'ALCOHOL',
          'OTHER', 'BENZODIAZEPINES (I.E. VALIUM, XANAX, ATIVAN, ETC)',
          'COCAINE/CRACK', 'PHARMACEUTICAL OPIOID', 'METHAMPHETAMINE',
          'SYNTHETIC MARIJUANA', 'PHARMACEUTICAL STIMULANT',
          'BARBITURATES (I.E. AMYTAL, NEMBUTAL, ETC)',
          'BUPRENORPHINE (I.E. SUBOXONE, SUBLOCADE, SUBUTEX, BUTRANS, BUPRENEX,
          ETC)',
          'XYLAZINE', 'BATH SALTS'], dtype=object)
```

```
[ ]: susp_drugs = ['HEROIN', 'FENTANYL', 'CARFENTANIL',
                  'FENTANYL ANALOG/OTHER SYNTHETIC OPIOID', 'METHADONE',
                  'PHARMACEUTICAL OPIOID',
                  'BUPRENORPHINE (I.E. SUBOXONE, SUBLOCADE, SUBUTEX, BUTRANS, BUPRENEX,
                  ETC)']
```

```
[ ]: down_sel2 = down_sel[down_sel['susp_od_drug_desc'].isin(susp_drugs)].copy()
```

```
[ ]: down_sel2
```

```
[ ]:
      dose_count  incident_id incident_time      day incident_county_name \
0              1          184    13:52:00  Wednesday      Allegheny
1              2         39237     4:13:00   Saturday      Allegheny
4              1         25677    17:30:00  Wednesday      Allegheny
6              1          3984     0:02:00   Friday       Allegheny
22             0         42561     8:00:00   Monday       Allegheny
...           ...           ...           ...           ...           ...
53879           0         40768    17:08:00   Tuesday      Allegheny
53891           2         37405    16:00:00   Friday       Allegheny
53904           2         39762    23:00:00   Monday       Allegheny
53929           1         39064    21:16:00   Sunday       Allegheny
53934           1         12293    21:30:00   Monday       Allegheny

      incident_state  victim_id  gender_desc  age_range  race  ... \
0      Pennsylvania      124      Female    20 - 24  WHITE  ...
1      Pennsylvania     31680      Female    50 - 59  WHITE  ...
4      Pennsylvania     20086       Male    50 - 59  WHITE  ...
6      Pennsylvania      3252       Male    50 - 59  WHITE  ...
22     Pennsylvania     34747       Male    40 - 49  WHITE  ...
...           ...           ...           ...           ...           ...
53879     Pennsylvania     33077       Male    30 - 39  WHITE  ...
53891     Pennsylvania     30037       Male    30 - 39  WHITE  ...
53904     Pennsylvania     32155       Male    30 - 39  WHITE  ...
53929     Pennsylvania     31523       Male    40 - 49  WHITE  ...
```

53934 Pennsylvania 9750 Female 30 - 39 WHITE ...

	geocoded_column	geocoded_column_address \
0	POINT (-79.986198 40.467355)	NaN
1	POINT (-79.986198 40.467355)	NaN
4	POINT (-79.986198 40.467355)	NaN
6	POINT (-79.986198 40.467355)	NaN
22	POINT (-79.986198 40.467355)	NaN
...	...	...
53879	POINT (-79.986198 40.467355)	NaN
53891	POINT (-79.986198 40.467355)	NaN
53904	POINT (-79.986198 40.467355)	NaN
53929	POINT (-79.986198 40.467355)	NaN
53934	POINT (-79.986198 40.467355)	NaN

	geocoded_column_city	geocoded_column_zip	geocoded_column_1_city \
0	NaN	NaN	NaN
1	NaN	NaN	NaN
4	NaN	NaN	NaN
6	NaN	NaN	NaN
22	NaN	NaN	NaN
...	...	...	...
53879	NaN	NaN	NaN
53891	NaN	NaN	NaN
53904	NaN	NaN	NaN
53929	NaN	NaN	NaN
53934	NaN	NaN	NaN

	geocoded_column_1_zip	geocoded_column_1_state \
0	NaN	NaN
1	NaN	NaN
4	NaN	NaN
6	NaN	NaN
22	NaN	NaN
...	...	...
53879	NaN	NaN
53891	NaN	NaN
53904	NaN	NaN
53929	NaN	NaN
53934	NaN	NaN

	geocoded_column_1	dose_unit	geocoded_column_1_address
0	POINT (-79.986198 40.467355)	0	NaN
1	POINT (-79.986198 40.467355)	0	NaN
4	POINT (-79.986198 40.467355)	0	NaN
6	POINT (-79.986198 40.467355)	0	NaN
22	POINT (-79.986198 40.467355)	0	NaN

```

...
53879 POINT (-79.917118 40.910832) ... 0 NaN
53891 POINT (-79.986198 40.467355) ... 4 NaN
53904 POINT (-79.986198 40.467355) ... 4 NaN
53929 POINT (-79.986198 40.467355) ... 4 NaN
53934 POINT (-79.986198 40.467355) ... 0 NaN

```

[5313 rows x 42 columns]

```
[ ]: # Use this cell for dataset information. Just change the column name
down_sel2['naloxone_administered'].value_counts()
```

```
[ ]: naloxone_administered
Y    4115
N    1198
Name: count, dtype: int64
```

```
[ ]: print(down_sel2['incident_county_name'].unique())
# print(down_sel2['incident_time'].unique())
print(down_sel2['day'].unique())
print(down_sel2['gender_desc'].unique())
print(down_sel2['age_range'].unique())
print(down_sel2['ethnicity_desc'].unique())
print(down_sel2['accidental_exposure'].unique())
print(down_sel2['susp_od_drug_desc'].unique())
print(down_sel2['incident_date'].unique())
print(down_sel2['response_time_desc'].unique())
print(down_sel2['survive'].unique())
print(down_sel2['incident_county_lat'].unique())
print(down_sel2['incident_county_long'].unique())
print(down_sel2['naloxone_administered'].unique())
```

```

['Allegheny']
['Wednesday' 'Saturday' 'Friday' 'Monday' 'Sunday' 'Thursday' 'Tuesday']
['Female', 'Male', 'Unknown']
Categories (3, object): ['Female', 'Male', 'Unknown']
['20 - 24' '50 - 59' '40 - 49' '30 - 39' '25 - 29' '70 - 79' '15 - 19'
 '60 - 69' '0 - 9' '14-Oct' '80 - *']
['Unknown' 'NON-HISPANIC OR NOT LATINO' 'HISPANIC or LATINO']
['N' 'Y']
['HEROIN' 'FENTANYL' 'CARFENTANIL'
 'FENTANYL ANALOG/OTHER SYNTHETIC OPIOID' 'METHADONE'
 'PHARMACEUTICAL OPIOID'
 'BUPRENORPHINE (I.E. SUBOXONE, SUBLOCADE, SUBUTEX, BUTRANS, BUPRENEX, ETC)']
['3/7/2018' '4/22/2023' '5/26/2021' ... '12/20/2019' '11/9/2019'
 '12/23/2022']
[nan '1-3 MINUTES' '<1 MINUTE' 'DID NOT WORK' '>5 MINUTES' '3-5 MINUTES'
 "DON'T KNOW"]

```

```
['N' 'U' 'Y']
[40.467355]
[-79.986198]
['Y' 'N']
```

```
[ ]: down_sel2.dtypes
```

```
[ ]: dose_count                int64
    incident_id                int64
    incident_time              object
    day                        object
    incident_county_name       object
    incident_state             object
    victim_id                  int64
    gender_desc                category
    age_range                  object
    race                       object
    ethnicity_desc             object
    victim_state              object
    victim_county             object
    accidental_exposure        object
    victim_od_drug_id          int64
    susp_od_drug_desc          object
    naloxone_administered      object
    administration_id          int64
    incident_date              object
    dose_desc                  object
    response_time_desc         object
    survive                   object
    response_desc              object
    revive_action_desc         object
    third_party_admin_desc     object
    incident_county_fips_code   int64
    incident_county_lat        float64
    incident_county_long       float64
    victim_county_fips_code     float64
    victim_county_lat          float64
    victim_county_long         float64
    geocoded_column_state      float64
    geocoded_column            object
    geocoded_column_address    float64
    geocoded_column_city       float64
    geocoded_column_zip        float64
    geocoded_column_1_city     float64
    geocoded_column_1_zip      float64
    geocoded_column_1_state    float64
    geocoded_column_1          object
```

```
dose_unit          int64
geocoded_column_1_address float64
dtype: object
```

```
[ ]: type(down_sel2['incident_date'][0])
```

```
[ ]: str
```

```
[ ]: down_sel2['incident_time'][0]
```

```
[ ]: '13:52:00'
```

```
[ ]: down_sel2['gender_desc'] = down_sel2['gender_desc'].astype('category')
```

```
[ ]:
```

```
[ ]: type(down_sel2['gender_desc'][0])
```

```
[ ]: str
```

```
[ ]: down_sel2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5313 entries, 0 to 53934
Data columns (total 42 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   dose_count                            5313 non-null   int64
1   incident_id                           5313 non-null   int64
2   incident_time                         5313 non-null   object
3   day                                   5313 non-null   object
4   incident_county_name                  5313 non-null   object
5   incident_state                        5313 non-null   object
6   victim_id                             5313 non-null   int64
7   gender_desc                           5313 non-null   category
8   age_range                             5313 non-null   object
9   race                                  5313 non-null   object
10  ethnicity_desc                        5313 non-null   object
11  victim_state                          4921 non-null   object
12  victim_county                         4921 non-null   object
13  accidental_exposure                   5313 non-null   object
14  victim_od_drug_id                     5313 non-null   int64
15  susp_od_drug_desc                     5313 non-null   object
16  naloxone_administered                 5313 non-null   object
17  administration_id                     5313 non-null   int64
18  incident_date                         5313 non-null   object
19  dose_desc                             4115 non-null   object
20  response_time_desc                    3835 non-null   object
```

```

21 survive 5313 non-null object
22 response_desc 4118 non-null object
23 revive_action_desc 3835 non-null object
24 third_party_admin_desc 2116 non-null object
25 incident_county_fips_code 5313 non-null int64
26 incident_county_lat 5313 non-null float64
27 incident_county_long 5313 non-null float64
28 victim_county_fips_code 4921 non-null float64
29 victim_county_lat 5313 non-null float64
30 victim_county_long 5313 non-null float64
31 geocoded_column_state 0 non-null float64
32 geocoded_column 5313 non-null object
33 geocoded_column_address 0 non-null float64
34 geocoded_column_city 0 non-null float64
35 geocoded_column_zip 0 non-null float64
36 geocoded_column_1_city 0 non-null float64
37 geocoded_column_1_zip 0 non-null float64
38 geocoded_column_1_state 0 non-null float64
39 geocoded_column_1 5313 non-null object
40 dose_unit 5313 non-null int64
41 geocoded_column_1_address 0 non-null float64
dtypes: category(1), float64(13), int64(7), object(21)
memory usage: 1.8+ MB

```

[]: